

Initiating Search

October 13, 2025, 8:29 PM

•• Search: Substitution of Aromatic Halides with Nitrogen Nucleophiles

Filtered By:

Yield: 90-100%, 80-89%, 70-79%, <10%
Document Type: Journal
Language: English
Publication Year: 2023 to 2025
Publication Name: Journal of Medicinal Chemistry, Journal of Organic Chemistry, Organic Letters, European Journal of Medicinal Chemistry, Journal of the American Chemical Society, Angewandte Chemie, International Edition, Bioorganic & Medicinal Chemistry Letters, Organic Chemistry Frontiers, Chemical Science, Green Chemistry, Bioorganic & Medicinal Chemistry, Nature Communications, ACS Medicinal Chemistry Letters, Nature (London, United Kingdom), Medicinal Chemistry Research

Search Tasks

Task	Result Type	View
Exported: Returned Reaction Results + Filters (9,310)	 Reactions	View Results

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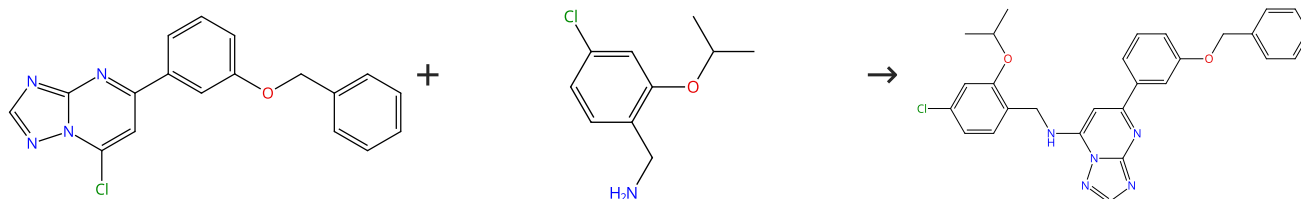
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Reactions (100)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (6)

31-614-CAS-38198292

Steps: 1 Yield: 100%

Development of Fluorescent AF64394 Analogues Enables Real-Time Binding Studies for the Orphan Class A GPCR GPR3

By: Bresinsky, Merlin; et al

Journal of Medicinal Chemistry (2023), 66(21), 15025-15041.

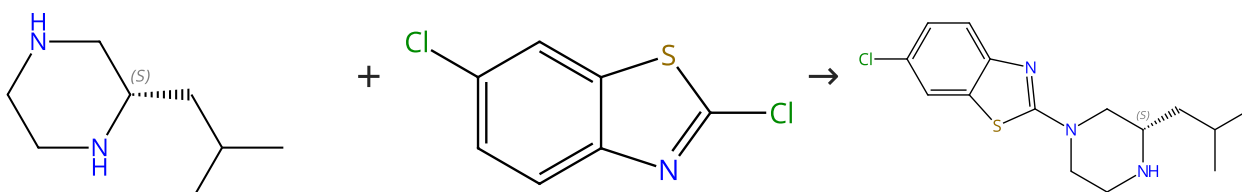
1.1 Reagents: Triethylamine

Solvents: Dichloromethane; 2 d, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

 Suppliers (25)

 Suppliers (87)

31-614-CAS-36251525

Steps: 1 Yield: 100%

Discovery and structure-activity relationship study of 2-piperazinyl-benzothiazole derivatives as potent and selective PPARδ agonists

By: Kato, Terukazu; et al

Bioorganic & Medicinal Chemistry (2023), 82, 117215.

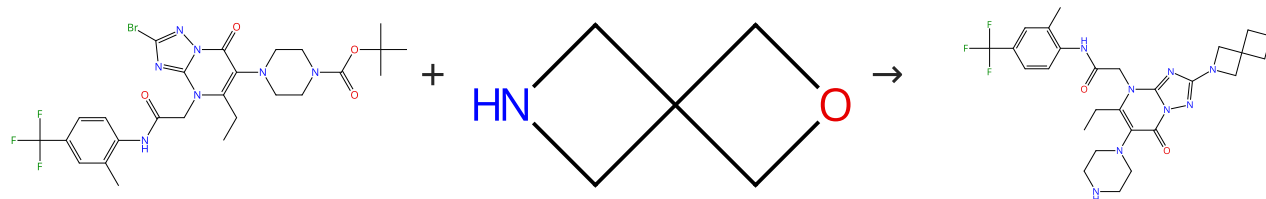
1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 12 h, rt

Experimental Protocols

Scheme 3 (1 Reaction)

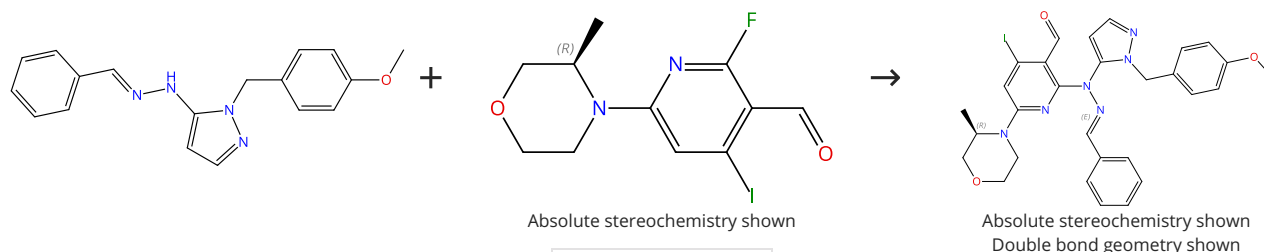
Steps: 1 Yield: 100%


 Suppliers (79)

31-614-CAS-42835785	Steps: 1 Yield: 100%	Design, synthesis, and structure-activity relationship studies of triazolo-pyrimidine derivatives as WRN inhibitors for the treatment of MSI tumors
1.1 Reagents: Potassium acetate Solvents: Dimethyl sulfoxide; 12 h, 120 °C		By: Sui, Qibang; et al
1.2 Reagents: Trifluoroacetic acid Solvents: Methanol, Dichloromethane; 10 min, rt		European Journal of Medicinal Chemistry (2025), 282, 117039.
Experimental Protocols		

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%

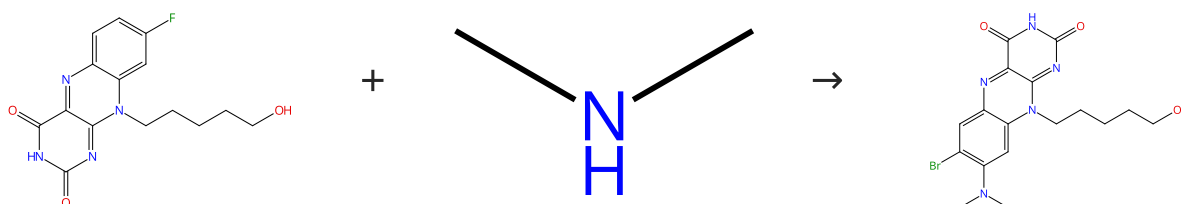


Suppliers (10)

31-614-CAS-39174871	Steps: 1 Yield: 100%	Discovery of the potent and selective ATR inhibitor camonsertib (RP-3500)
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; 15 min, rt; < 26 °C; 0.5 h, < 26 °C		By: Black, W. Cameron; et al
1.2 < 26 °C; 0.5 h, < 26 °C		Journal of Medicinal Chemistry (2024), 67(4), 2349-2368.
Experimental Protocols		

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%

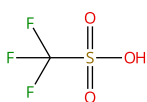


Suppliers (118)

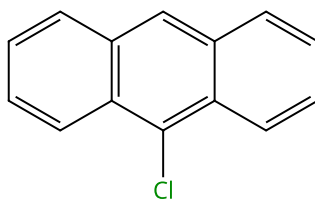
31-614-CAS-35674132	Steps: 1 Yield: 100%	Enhancing the photosensitizing activity of natural flavins: Tuning the heavy-atom effect in the isoalloxazine series
1.1 Solvents: Dimethylformamide, Tetrahydrofuran; 12 h, rt		By: Ribes, Jonathan; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2023), 81, 117210.

Scheme 6 (1 Reaction)

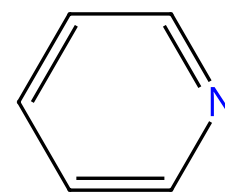
Steps: 1 Yield: 100%



+



+



→

• 1/3 Sc(III)

Suppliers (62)

Suppliers (63)

Suppliers (211)

Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-46428633

Steps: 1 Yield: 100%

Visible-Light S_NAr Pyridination of Aryl Halides and Triflates with Acridine-Lewis Acid Photocatalysts

By: Reich, Sabrina M.; et al

Journal of Organic Chemistry (2025), 90(29), 10332-10337.

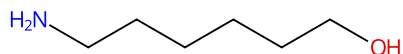
1.1 Catalysts: Acridine

Solvents: 2,2,2-Trifluoroethanol, 1,2-Dichloroethane; 24 h, rt

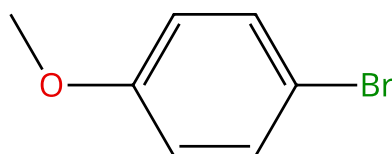
Experimental Protocols

Scheme 7 (1 Reaction)

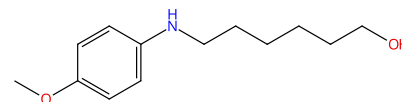
Steps: 1 Yield: 100%



+



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Suppliers (99)

Suppliers (71)

Supplier (1)

31-614-CAS-46331819

Steps: 1 Yield: 100%

Efficient Aminations of Aryl Halides by a Cu(II) Catalyst

By: Pierson, Christina N.; et al

Journal of the American Chemical Society (2025), 147(24), 20939-20946.

1.1 Reagents: Tripotassium phosphate

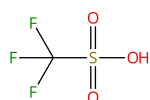
Catalysts: Copper bromide (CuBr_2), N^1,N^2 -Bis(2,5-dimethyl-1H-pyrrol-1-yl)ethanediamide

Solvents: Dimethyl sulfoxide; 24 h, 100 °C

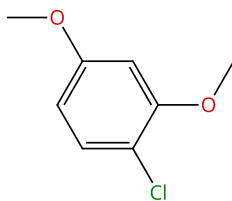
Experimental Protocols

Scheme 8 (1 Reaction)

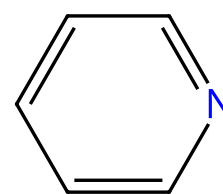
Steps: 1 Yield: 100%



+



+



→

• 1/3 Sc(III)

Suppliers (62)

Suppliers (55)

Suppliers (211)

Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-46428630

Steps: 1 Yield: 100%

Visible-Light S_NAr Pyridination of Aryl Halides and Triflates with Acridine-Lewis Acid Photocatalysts

By: Reich, Sabrina M.; et al

Journal of Organic Chemistry (2025), 90(29), 10332-10337.

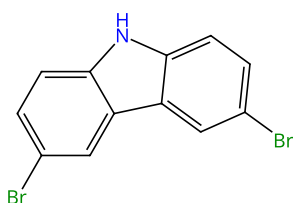
1.1 Catalysts: Acridine

Solvents: 2,2,2-Trifluoroethanol, 1,2-Dichloroethane; 24 h, rt

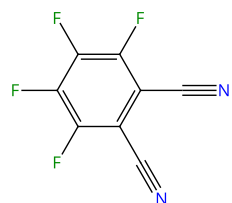
Experimental Protocols

Scheme 9 (1 Reaction)

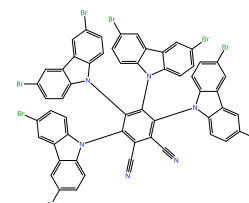
Steps: 1 Yield: 100%



+



→



Suppliers (93)

Suppliers (76)

31-614-CAS-41316037

Steps: 1 Yield: 100%

Radical Deoxygenative Three-Component Reaction of Alcohols, Aryl Alkenes, and Cyanopyridines

By: Feng, Li-Min; et al

Organic Letters (2024), 26(29), 6225-6229.

1.1 Reagents: Sodium hydride

Solvents: Tetrahydrofuran; rt; 30 min, rt

1.2 12 h, rt

1.3 Reagents: Water

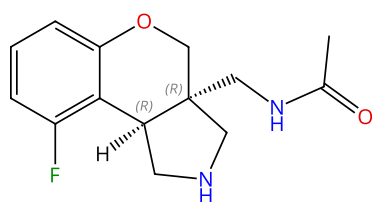
1.4 Reagents: Sodium bicarbonate

Solvents: Water; neutralized

Experimental Protocols

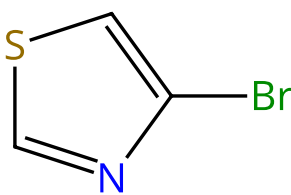
Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%

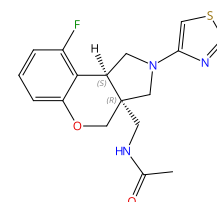


Relative stereochemistry shown

+



→



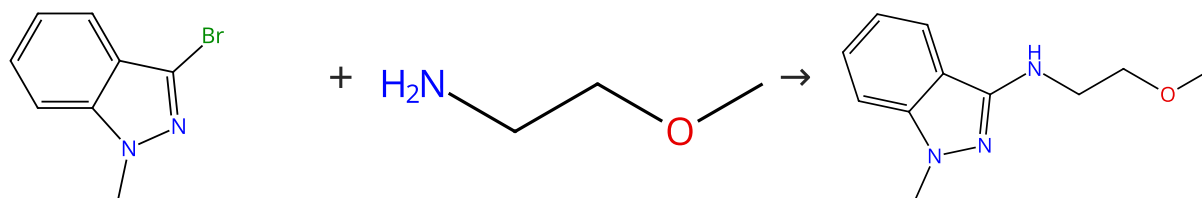
Relative stereochemistry shown

Suppliers (97)

31-614-CAS-35422568 Steps: 1 Yield: 100% 1.1 Reagents: Sodium trimethylsilanolate Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl- κC^1]phosphine- κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (<i>SP</i> -4-2)- Solvents: Tetrahydrofuran; rt $\rightarrow$ 50 °C; 3 h, 50 °C	Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines By: Reichert, Elaine C.; et al Journal of the American Chemical Society (2023), 145(6), 3323-3329.
Experimental Protocols	

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



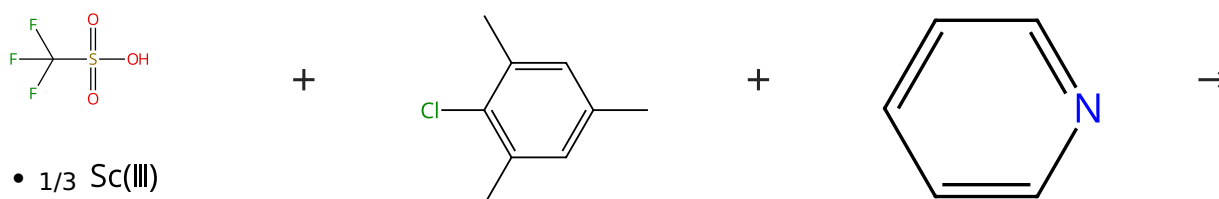
Suppliers (60)

Suppliers (60)

31-614-CAS-35422549 Steps: 1 Yield: 100% 1.1 Reagents: Sodium trimethylsilanolate Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl- κC^1]phosphine- κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (<i>SP</i> -4-2)- Solvents: Tetrahydrofuran; 3 h, 90 °C	Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines By: Reichert, Elaine C.; et al Journal of the American Chemical Society (2023), 145(6), 3323-3329.
Experimental Protocols	

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (62)

Suppliers (60)

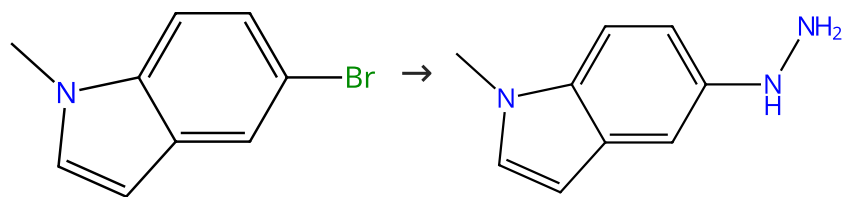
Suppliers (211)

Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-46428635 Steps: 1 Yield: 100%	Visible-Light S_NAr Pyridination of Aryl Halides and Triflates with Acridine-Lewis Acid Photocatalysts
1.1 Catalysts: Acridine Solvents: 2,2,2-Trifluoroethanol, 1,2-Dichloroethane; 24 h, rt	By: Reich, Sabrina M.; et al
Experimental Protocols	Journal of Organic Chemistry (2025), 90(29), 10332-10337.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (82)

Suppliers (5)

31-614-CAS-46331869

Steps: 1 Yield: 100%

Efficient Aminations of Aryl Halides by a Cu(II) Catalyst

1.1 **Reagents:** Hydrazine, Tripotassium phosphate
Catalysts: Copper bromide (CuBr₂), *N*¹,*N*²-Bis(2,5-dimethyl-1*H*-pyrrol-1-yl)ethanediamide
Solvents: Dimethyl sulfoxide; 24 h, 65 °C

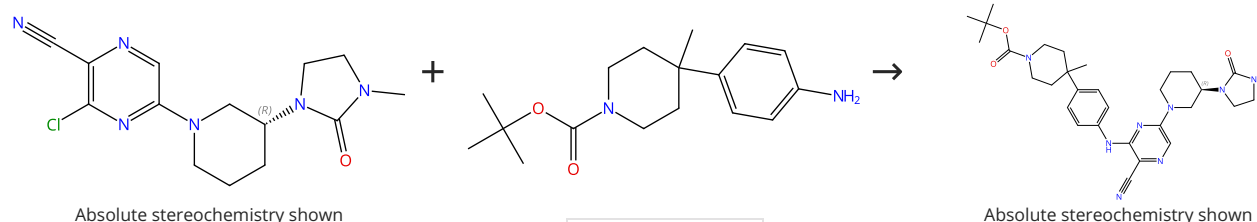
By: Pierson, Christina N.; et al

Journal of the American Chemical Society (2025), 147(24), 20939-20946.

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (13)

Supplier (1)

31-614-CAS-40809771

Steps: 1 Yield: 100%

Discovery and Preclinical Pharmacology of NX-2127, an Orally Bioavailable Degradable of Bruton's Tyrosine Kinase with Immunomodulatory Activity for the Treatment of Patients with B Cell Malignancies

1.1 **Reagents:** Cesium carbonate
Catalysts: Palladium diacetate, BINAP
Solvents: 1,4-Dioxane; 2 h, 115 °C

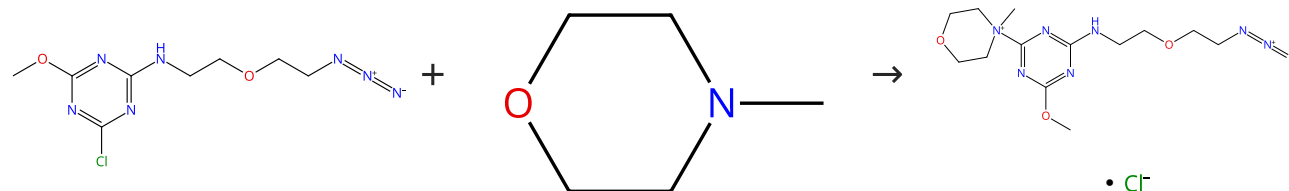
By: Robbins, Daniel W.; et al

Journal of Medicinal Chemistry (2024), 67(4), 2321-2336.

Experimental Protocols

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%

• Cl⁻

Suppliers (81)

31-614-CAS-40486507

Steps: 1 Yield: 100%

Selective Arylation of RNA 2'-OH Groups via S_NAr Reaction with Trialkylammonium Heterocycles1.1 **Solvents:** Tetrahydrofuran; 30 min, 25 °C

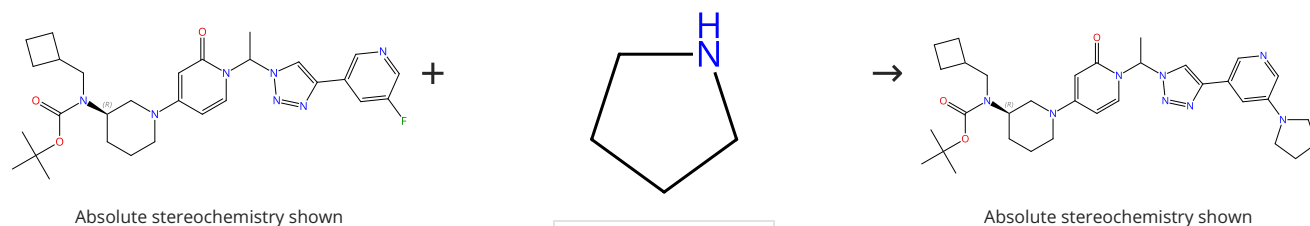
By: Chatterjee, Sayantan; et al

Angewandte Chemie, International Edition (2024), 63(25), e202403496.

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

31-614-CAS-44468579

Steps: 1 Yield: 100%

1.1 **Reagents:** Diisopropylethylamine
Solvents: *N*-Methyl-2-pyrrolidone; 20 h, 120 °C

Experimental Protocols

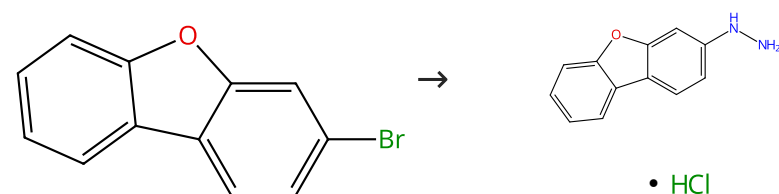
Discovery, Optimization, and Preclinical Pharmacology of E P652, a METTL3 Inhibitor with Efficacy in Liquid and Solid Tumor Models

By: Dutheuil, Guillaume; et al

Journal of Medicinal Chemistry (2025), 68(3), 2981-3003.

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (66)

Suppliers (3)

31-614-CAS-46331852

Steps: 1 Yield: 100%

1.1 **Reagents:** Hydrazine, Tripotassium phosphate
Catalysts: Copper bromide (CuBr₂), *N*¹,*N*²-Bis(2,5-dimethyl-1*H*-pyrrol-1-yl)ethanediamide
Solvents: Dimethyl sulfoxide; 24 h, 65 °C

1.2 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

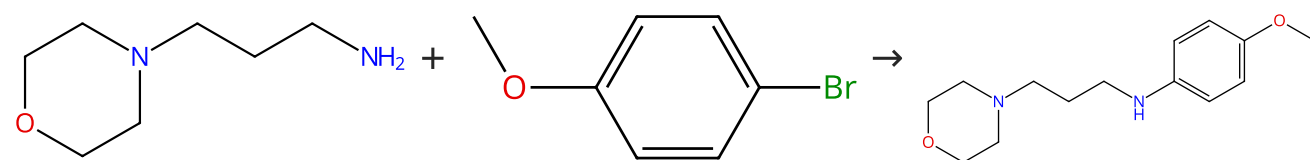
Efficient Aminations of Aryl Halides by a Cu(II) Catalyst

By: Pierson, Christina N.; et al

Journal of the American Chemical Society (2025), 147(24), 20939-20946.

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

Suppliers (71)

Supplier (1)

31-614-CAS-46331810

Steps: 1 Yield: 100%

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Copper bromide (CuBr₂), *N*¹,*N*²-Bis(2,5-dimethyl-1*H*-pyrrol-1-yl)ethanediamide
Solvents: Dimethyl sulfoxide; 24 h, 100 °C

Experimental Protocols

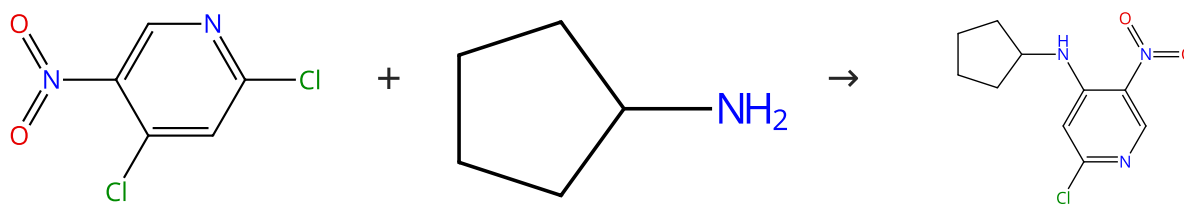
Efficient Aminations of Aryl Halides by a Cu(II) Catalyst

By: Pierson, Christina N.; et al

Journal of the American Chemical Society (2025), 147(24), 20939-20946.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (91)

Suppliers (63)

Supplier (1)

31-614-CAS-41273748

Steps: 1 Yield: 100%

Identification of 6-Anilino Imidazo[4,5-c]pyridin-2-ones as Selective DNA-Dependent Protein Kinase Inhibitors and Their Application as Radiosensitizers

1.1 Reagents: Diisopropylethylamine
Solvents: Dichloromethane; 5 °C; 16 h, 20 °C

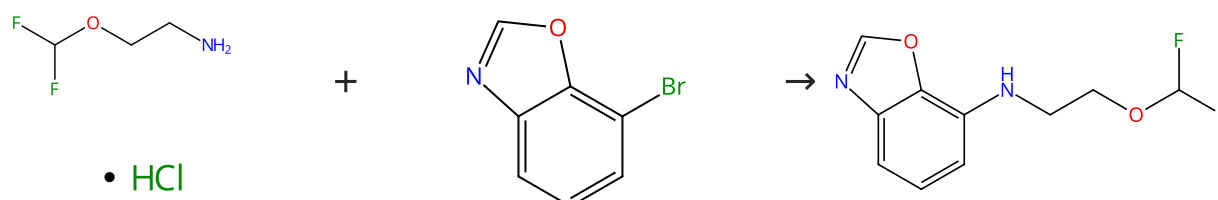
Experimental Protocols

By: Hong, Cho R.; et al

Journal of Medicinal Chemistry (2024), 67(14), 12366-12385.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



• HCl

Suppliers (26)

Suppliers (69)

31-614-CAS-41352359

Steps: 1 Yield: 100%

Discovery of BI-9508, a Brain-Penetrant GPR88-Receptor-Agonist Tool Compound for In Vivo Mouse Studies

1.1 Reagents: Cesium carbonate
Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]bis(1,1-dimethylethyl)phosphine-κP](methanesulfonato-κO) palladium
Solvents: 1,4-Dioxane; 2 h, 90 °C; cooled

1.2 Reagents: Water

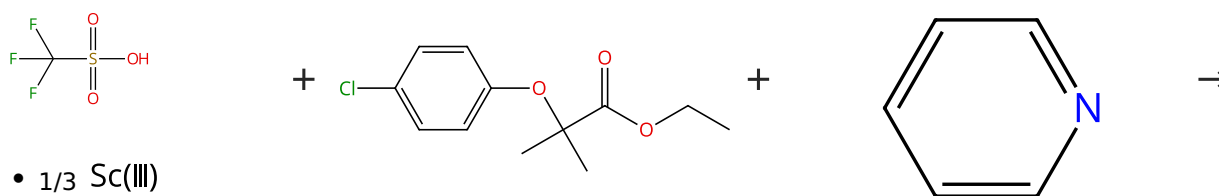
Experimental Protocols

By: Fer, Mickael; et al

Journal of Medicinal Chemistry (2024), 67(13), 11296-11325.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



• 1/3 Sc(III)

Suppliers (62)

Suppliers (87)

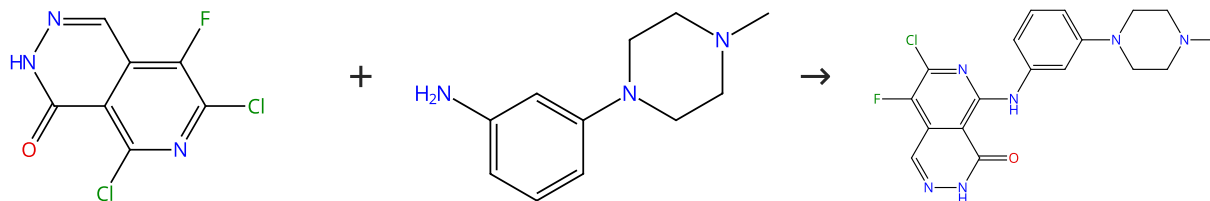
Suppliers (211)

Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-46428639	Steps: 1 Yield: 100%	Visible-Light S_NAr Pyridination of Aryl Halides and Triflates with Acridine-Lewis Acid Photocatalysts
1.1 Catalysts: Acridine Solvents: 2,2,2-Trifluoroethanol, 1,2-Dichloroethane; 24 h, rt		By: Reich, Sabrina M.; et al
Experimental Protocols		Journal of Organic Chemistry (2025), 90(29), 10332-10337.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



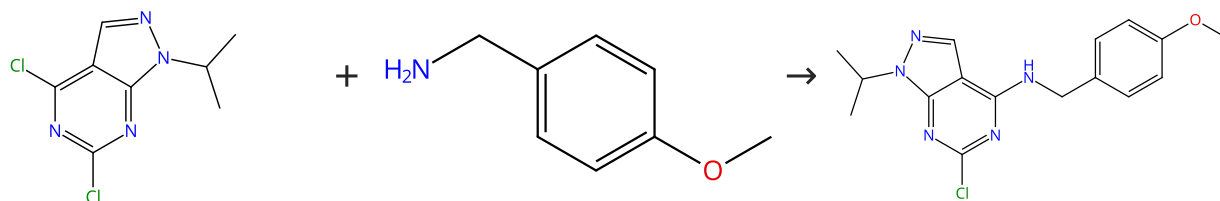
Suppliers (9)

Suppliers (78)

31-614-CAS-41130630	Steps: 1 Yield: 100%	Optimization of Novel Pyrido-pyridazinone Derivatives as FER Tyrosine Kinase Inhibitors, Leading to the Potent DS08701581
1.1 Solvents: 1,4-Dioxane; 4 h, 105 °C		By: Taniguchi, Toru; et al
1.2 Reagents: Ethanol		ACS Medicinal Chemistry Letters (2024), 15(7), 1010-1016.
Experimental Protocols		

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



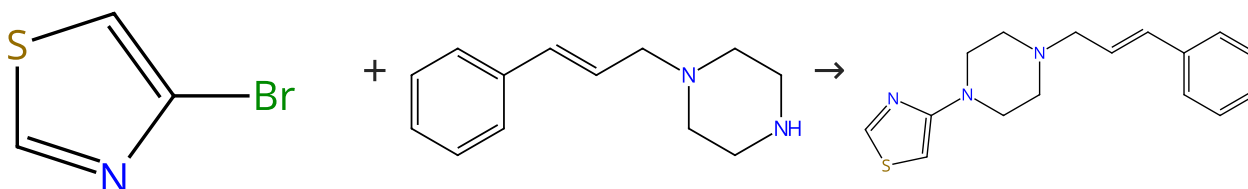
Suppliers (52)

Suppliers (91)

31-614-CAS-38257185	Steps: 1 Yield: 100%	Drug repurposing and structure-based discovery of new PDE4 and PDE5 inhibitors
1.1 Reagents: Diisopropylethylamine Solvents: Ethanol; 16 h, 60 °C		By: Liu, Jiayuan; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2023), 262, 115893.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



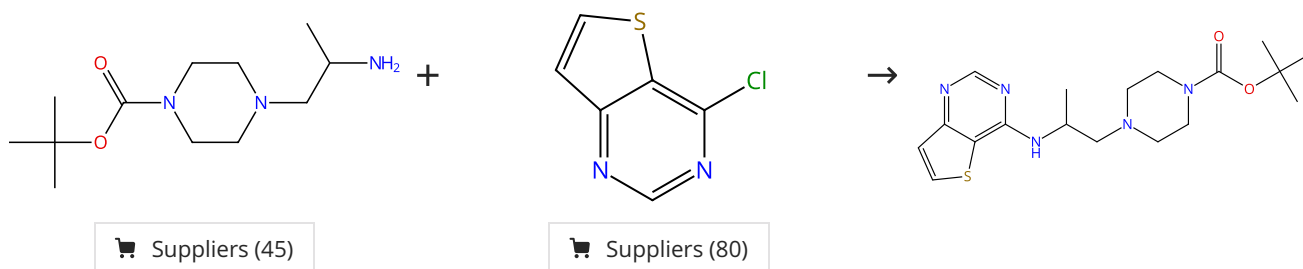
Suppliers (97)

Suppliers (42)

31-614-CAS-35422551 Steps: 1 Yield: 100% 1.1 Reagents: Sodium trimethylsilanolate Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC¹]]phosphine-κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (<i>SP</i>-4-2)- Solvents: Tetrahydrofuran; 3 h, 50 °C Experimental Protocols	Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines By: Reichert, Elaine C.; et al Journal of the American Chemical Society (2023), 145(6), 3323-3329.
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Scheme 25 (1 Reaction)

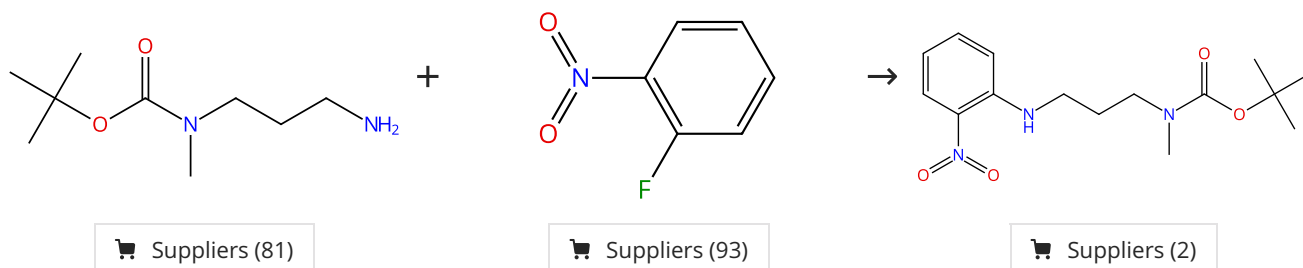
Steps: 1 Yield: 100%



31-614-CAS-40576919 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Solvents: Acetonitrile; 6 - 24 h, 85 °C Experimental Protocols	Discovery of a Potent, Selective, and Orally Bioavailable Tool Compound for Probing the Role of Lysophosphatidic Acid Type 2 Receptor Antagonists in Fibrotic Disorders By: Armani, Elisabetta; et al Journal of Medicinal Chemistry (2023), 66(8), 5622-5656.
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Scheme 26 (1 Reaction)

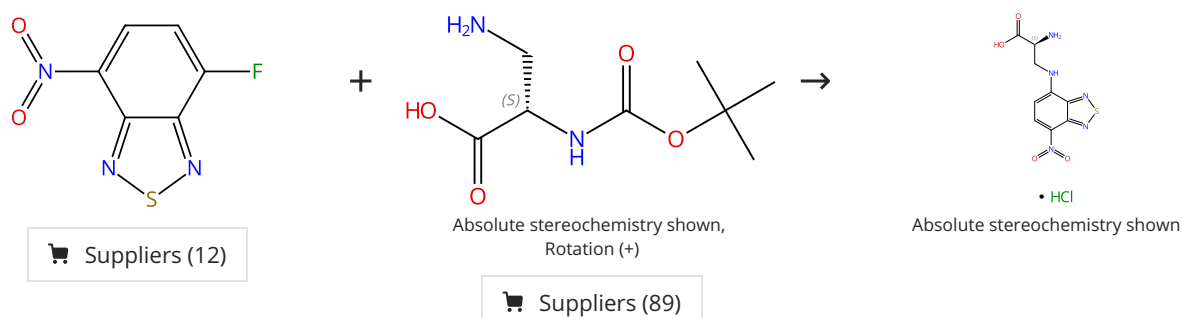
Steps: 1 Yield: 100%



31-614-CAS-39924806 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: Dimethylformamide; 20 h, 100 °C Experimental Protocols	Discovery and Characterization of a New Class of C5aR1 Antagonists Showing In Vivo Activity By: Hubler, Francis; et al Journal of Medicinal Chemistry (2024), 67(5), 4100-4119.
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Scheme 27 (1 Reaction)

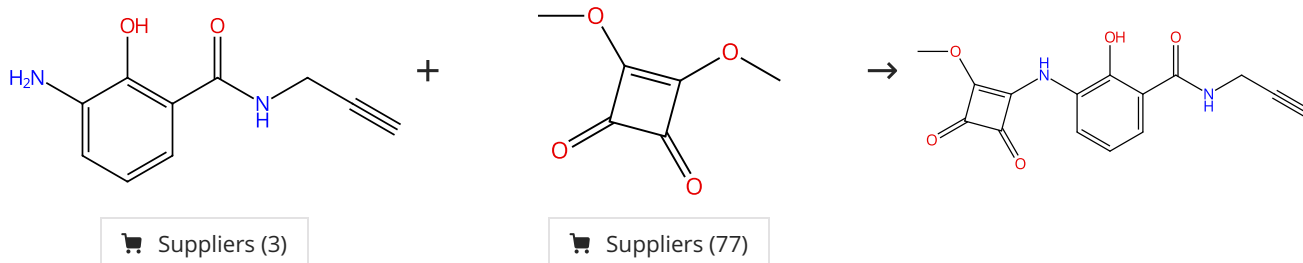
Steps: 1 Yield: 100%



31-614-CAS-39061016	Steps: 1 Yield: 100%	Small Fluorogenic Amino Acids for Peptide-Guided Background-Free Imaging By: de Moliner, Fabio; et al Angewandte Chemie, International Edition (2023), 62(4), e202216231.
1.1 Reagents: Sodium bicarbonate Solvents: Methanol, Water; 55 °C; 15 min, 55 °C		
1.2 Reagents: Hydrochloric acid Solvents: 1,4-Dioxane; 15 min, rt		
Experimental Protocols		

Scheme 28 (1 Reaction)

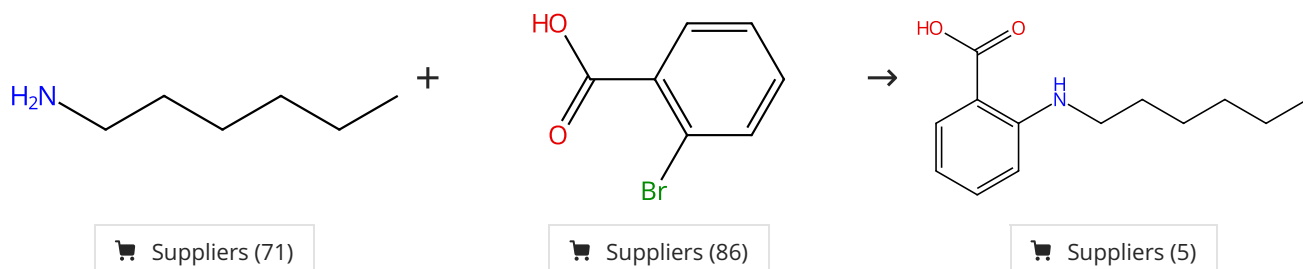
Steps: 1 Yield: 100%



31-614-CAS-44284136	Steps: 1 Yield: 100%	A Fluorescent Probe Enables the Discovery of Improved Antagonists Targeting the Intracellular Allosteric Site of the Chemokine Receptor CCR7 By: Wurnig, Silas L.; et al Journal of Medicinal Chemistry (2025), 68(4), 4308-4333.
1.1 Solvents: Methanol; 3 d, rt		

Scheme 29 (1 Reaction)

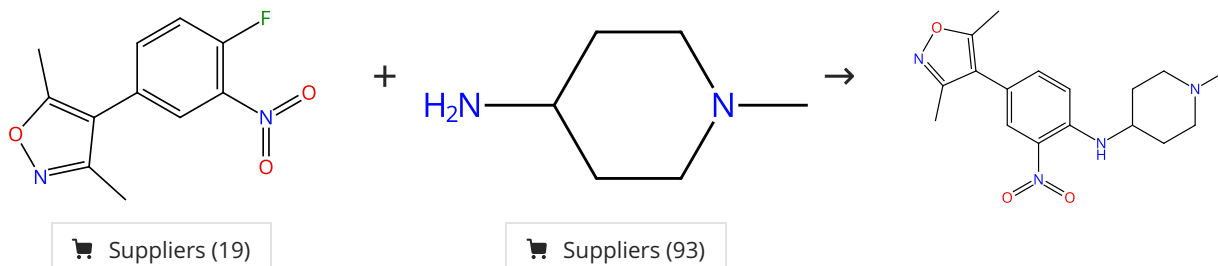
Steps: 1 Yield: 100%



31-614-CAS-46331797	Steps: 1 Yield: 100%	Efficient Aminations of Aryl Halides by a Cu(II) Catalyst By: Pierson, Christina N.; et al Journal of the American Chemical Society (2025), 147(24), 20939-20946.
1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (CuBr ₂), <i>N</i> ¹ , <i>N</i> ² -Bis(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 24 h, 100 °C		
Experimental Protocols		

Scheme 30 (1 Reaction)

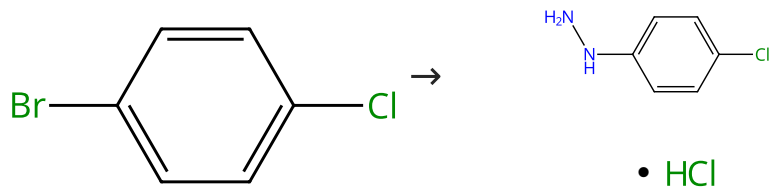
Steps: 1 Yield: 100%



31-614-CAS-40320918	Steps: 1 Yield: 100%	Discovery of Highly Potent and Efficient CBP/p300 Degraders with Strong In Vivo Antitumor Activity
1.1 Reagents: Triethylamine Solvents: Dimethylformamide; overnight, rt → 45 °C		By: Hu, Jiankang; et al
Experimental Protocols		Journal of Medicinal Chemistry (2024), 67(9), 6952-6986.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



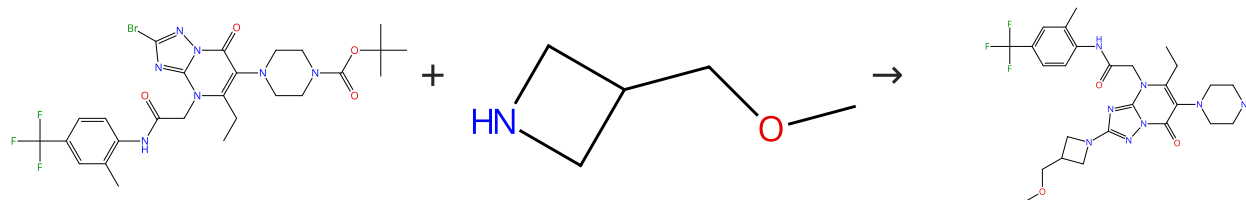
Suppliers (83)

Suppliers (93)

31-614-CAS-46331832	Steps: 1 Yield: 100%	Efficient Aminations of Aryl Halides by a Cu(II) Catalyst
1.1 Reagents: Hydrazine, Tripotassium phosphate Catalysts: Copper bromide (CuBr ₂), <i>N</i> ¹ , <i>N</i> ² -Bis(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 24 h, 65 °C		By: Pierson, Christina N.; et al
1.2 Reagents: Hydrochloric acid Solvents: Water		Journal of the American Chemical Society (2025), 147(24), 20939-20946.
Experimental Protocols		

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%

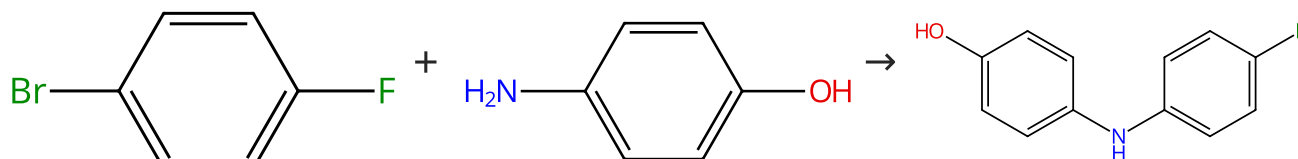


Suppliers (32)

31-614-CAS-42835783	Steps: 1 Yield: 100%	Design, synthesis, and structure-activity relationship studies of triazolo-pyrimidine derivatives as WRN inhibitors for the treatment of MSI tumors
1.1 Reagents: Potassium acetate Solvents: Dimethyl sulfoxide; 12 h, 120 °C		By: Sui, Qibang; et al
1.2 Reagents: Trifluoroacetic acid Solvents: Methanol, Dichloromethane; 10 min, rt		European Journal of Medicinal Chemistry (2025), 282, 117039.
Experimental Protocols		

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (103)

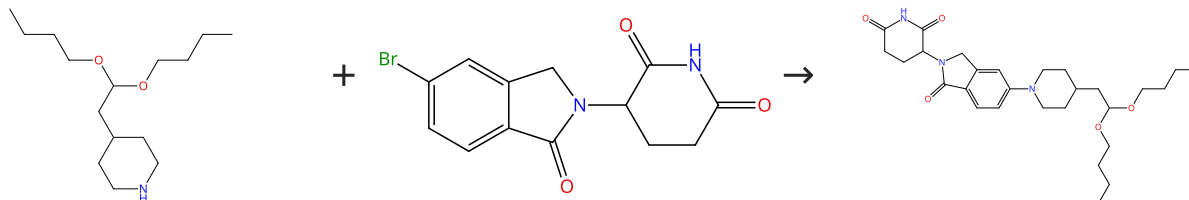
Suppliers (100)

Suppliers (4)

31-614-CAS-46330491 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Catalysts: Cuprous iodide, 2-Pyridinecarboxamide, <i>N</i>-(2,5-dimethyl-1<i>H</i>-pyrrol-1-yl)-1,6-dihydro-6-oxo- Solvents: Ethanol, Methanol; 24 h, 25 °C Experimental Protocols	Room-Temperature Copper Cross-Coupling Reactions of Anilines with Aryl Bromides By: Dueker, Jonas; et al Organic Letters (2025), 27(22), 5566-5571.
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Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%

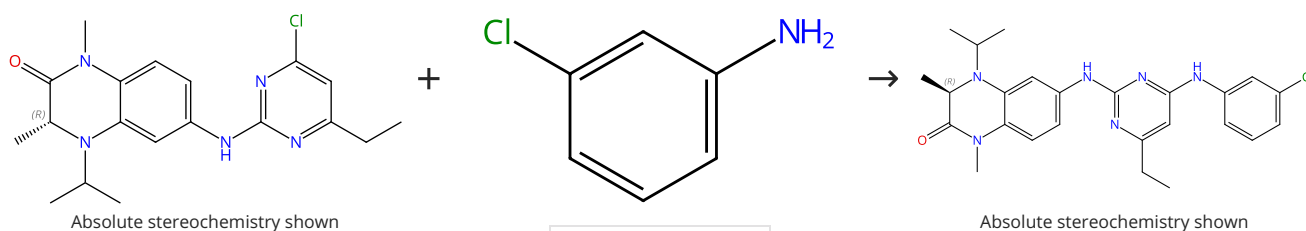


Suppliers (77)

31-614-CAS-38259517 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: (<i>SP</i>4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2<i>H</i>-imidazol-2-ylidene]dichloro(3-chloropyridine-κ<i>M</i>) palladium Solvents: 1,4-Dioxane; 20 °C; 3 h, 90 °C Experimental Protocols	Rapid PROTAC discovery platform: nanomole scale array synthesis and direct screening of reaction mixtures By: Plesniak, Mateusz P.; et al ACS Medicinal Chemistry Letters (2023), 14(12), 1882-1890.
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Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%

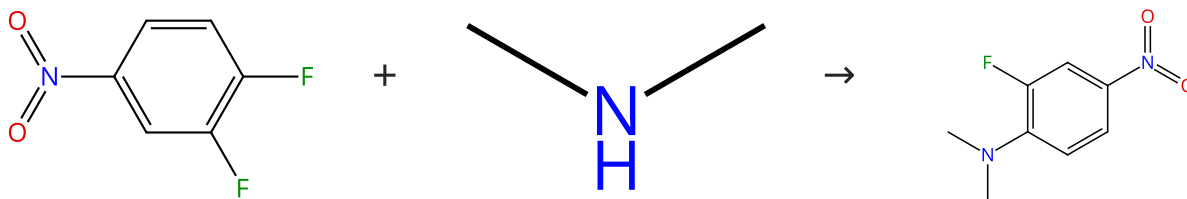


Suppliers (75)

31-614-CAS-44316188 Steps: 1 Yield: 100% 1.1 Reagents: Hydrochloric acid Solvents: <i>tert</i>-Butanol; 3 h, rt → 80 °C Experimental Protocols	Design of Selective BRD4 Inhibitors for the Treatment of Autosomal Dominant Polycystic Kidney Disease By: Yang, Yueyue; et al Journal of Medicinal Chemistry (2025), 68(5), 5257-5274.
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Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (88)

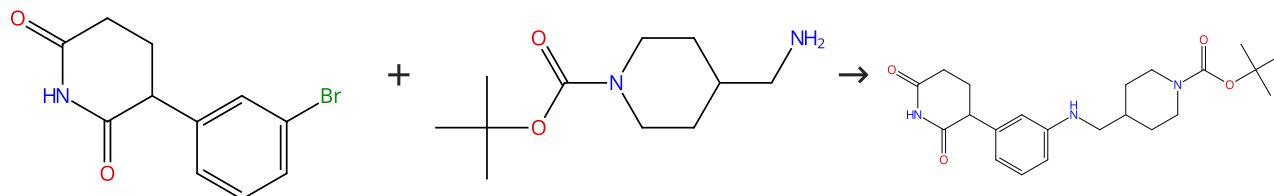
Suppliers (118)

Suppliers (41)

31-614-CAS-38403119	Steps: 1 Yield: 100%	Aryne-Enabled C-N Arylation of Anilines
1.1 Solvents: Tetrahydrofuran; rt; 16 h, rt		By: Sephton, Thomas; et al
1.2 Reagents: Water; rt		Angewandte Chemie, International Edition (2023), 62(49), e202310583.
Experimental Protocols		

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



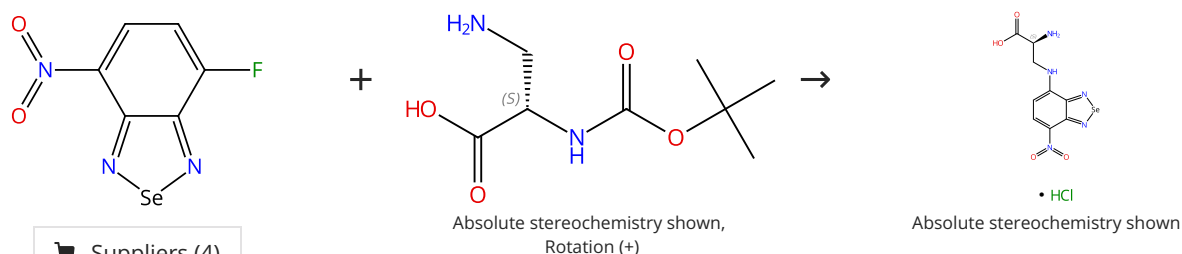
Suppliers (40)

Suppliers (98)

31-614-CAS-43333520	Steps: 1 Yield: 100%	Development of a Buchwald-Hartwig Amination for an Accelerated Library Synthesis of Cereblon Binders
1.1 Solvents: 1,4-Dioxane; 5 min, rt		By: Lejava, Anastasia; et al
1.2 Catalysts: Tris(dibenzylideneacetone)dipalladium, [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine Solvents: 1,4-Dioxane; 5 min, rt; 15 - 20 min, rt		ACS Medicinal Chemistry Letters (2025), 16(1), 89-95.
1.3 Reagents: Lithium bis(trimethylsilyl)amide Solvents: Toluene; 1 min, rt; 16 h, 45 °C; 45 °C → 0 °C		
1.4 Reagents: Citric acid Solvents: 1,4-Dioxane; 5 min, 0 °C		
Experimental Protocols		

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



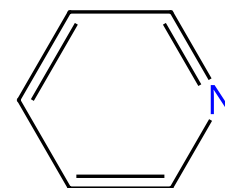
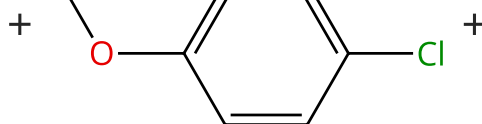
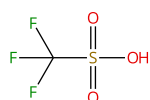
Suppliers (4)

Suppliers (89)

31-614-CAS-39061007	Steps: 1 Yield: 100%	Small Fluorogenic Amino Acids for Peptide-Guided Background-Free Imaging
1.1 Reagents: Sodium bicarbonate Solvents: Methanol, Water; 55 °C; 30 min, 55 °C		By: de Moliner, Fabio; et al
1.2 Reagents: Hydrochloric acid Solvents: 1,4-Dioxane; 30 min, rt		Angewandte Chemie, International Edition (2023), 62(4), e202216231.
Experimental Protocols		

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



• 1/3 Sc(III)

Suppliers (62)

Suppliers (73)

Suppliers (211)

Multi-component
structure image
available in CAS
SciFinder

31-614-CAS-46428615

Steps: 1 Yield: 100%

Visible-Light S_NAr Pyridination of Aryl Halides and Triflates
with Acridine-Lewis Acid Photocatalysts

1.1 Catalysts: Acridine

Solvents: 2,2,2-Trifluoroethanol, 1,2-Dichloroethane; 24 h, rt

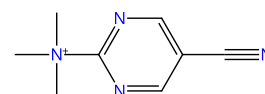
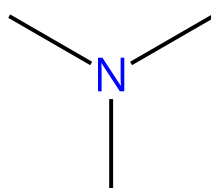
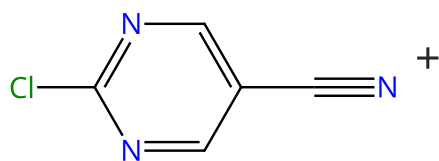
By: Reich, Sabrina M.; et al

Experimental Protocols

Journal of Organic Chemistry (2025), 90(29), 10332-10337.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 100%



• Cl⁻

Suppliers (69)

Suppliers (135)

31-614-CAS-40486509

Steps: 1 Yield: 100%

Selective Arylation of RNA 2'-OH Groups via S_NAr Reaction
with Trialkylammonium Heterocycles

1.1 Solvents: Tetrahydrofuran; 30 min, 25 °C

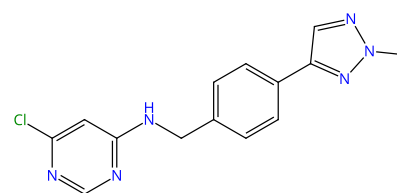
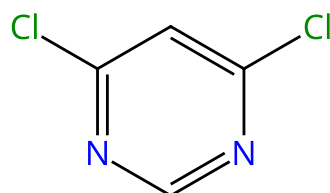
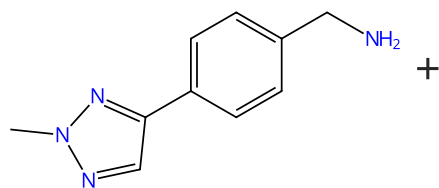
By: Chatterjee, Sayantan; et al

Experimental Protocols

Angewandte Chemie, International Edition (2024), 63(25), e202403496.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 100%



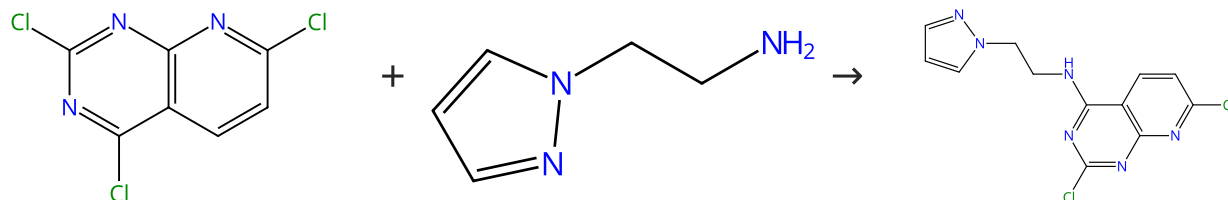
Suppliers (11)

Suppliers (94)

31-614-CAS-40478668	Steps: 1 Yield: 100%	Identification of M4205-A Highly Selective Inhibitor of KIT Mutations for Treatment of Unresectable Metastatic or Recurrent Gastrointestinal Stromal Tumors
1.1 Reagents: Triethylamine Solvents: Dimethylformamide; 2 h, 30 °C		By: Blum, Andreas; et al
Experimental Protocols		Journal of Medicinal Chemistry (2023), 66(4), 2386-2395.

Scheme 42 (1 Reaction)

Steps: 1 Yield: 100%



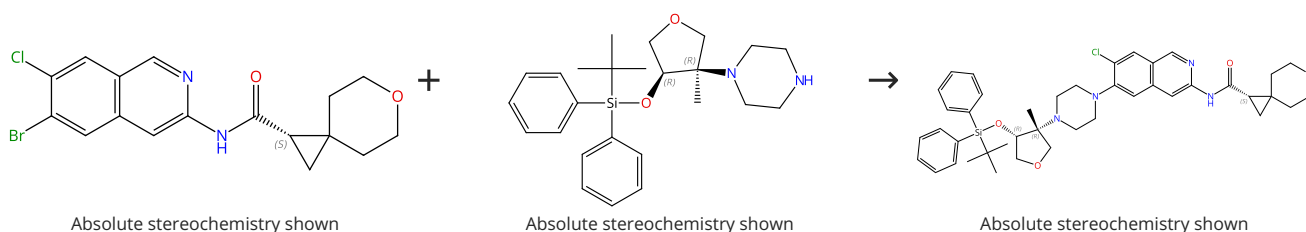
Suppliers (64)

Suppliers (74)

31-614-CAS-38181255	Steps: 1 Yield: 100%	Discovery and characterisation of quinazoles and 8-Azaquinazoles as NLRP3 agonists with oral bioavailability in mice
1.1 Reagents: Diisopropylethylamine Solvents: Dichloromethane; 0 °C → rt; 75 min, rt		By: O' Donovan, Daniel H.; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 96, 129518.

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

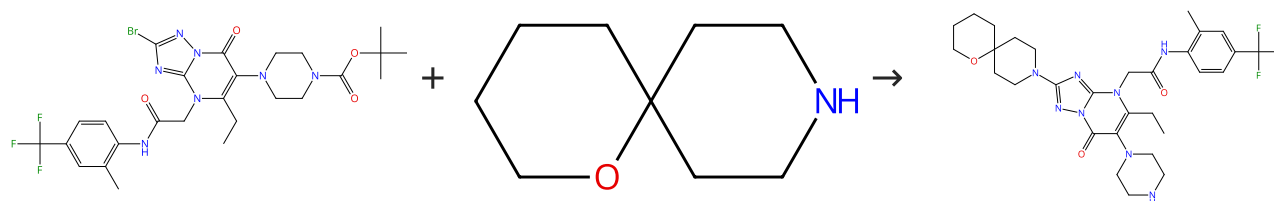
Absolute stereochemistry shown

Supplier (1)

31-614-CAS-38121374	Steps: 1 Yield: 100%	Discovery of MK-1468: A Potent, Kinome-Selective, Brain-Penetrant Amidoisoquinoline LRRK2 Inhibitor for the Potential Treatment of Parkinson's Disease
1.1 Catalysts: Di-μ-chlorobis(η ³ -2-propenyl)dipalladium, BINAP Solvents: Tetrahydrofuran; 10 min, rt		By: Kattar, Solomon D.; et al
1.2 Reagents: Sodium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; 4 h, rt → 60 °C		Journal of Medicinal Chemistry (2023), 66(21), 14912-14927.
Experimental Protocols		

Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%

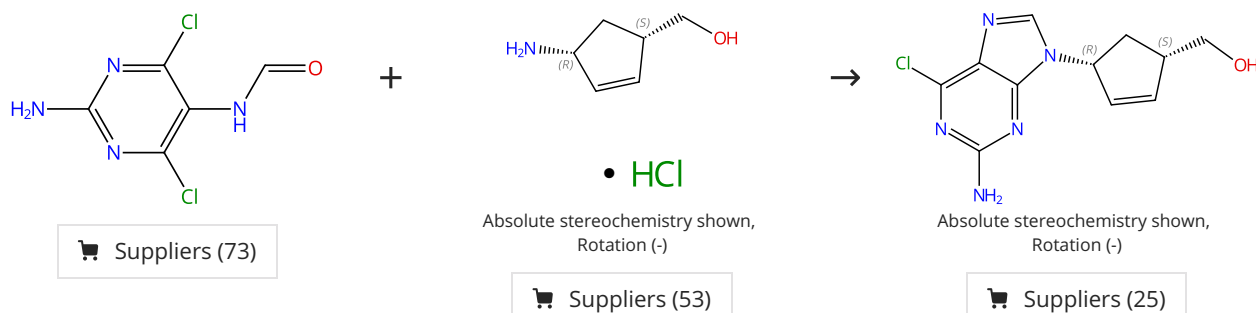


Suppliers (35)

31-614-CAS-42835786 Steps: 1 Yield: 100% 1.1 Reagents: Potassium acetate Solvents: Dimethyl sulfoxide; 12 h, 120 °C 1.2 Reagents: Trifluoroacetic acid Solvents: Methanol, Dichloromethane; 10 min, rt Experimental Protocols	Design, synthesis, and structure-activity relationship studies of triazolo-pyrimidine derivatives as WRN inhibitors for the treatment of MSI tumors By: Sui, Qibang; et al European Journal of Medicinal Chemistry (2025), 282, 117039.
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Scheme 45 (1 Reaction)

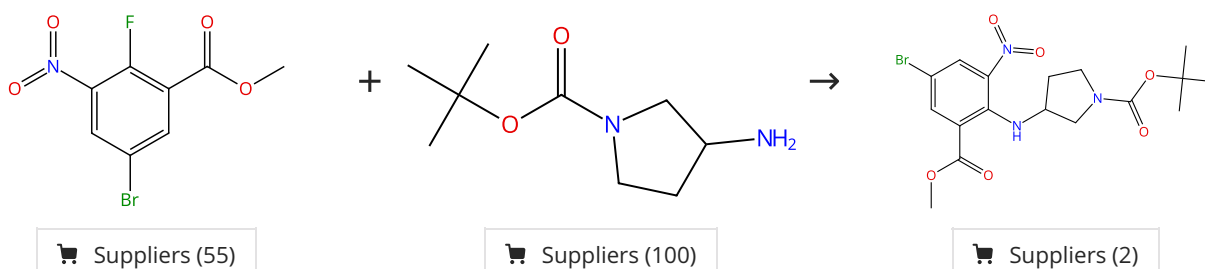
Steps: 1 Yield: 100%



31-614-CAS-39130578 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: Ethanol, Water; 1 h, 140 °C Experimental Protocols	Synthesis and Biophysical Properties of Phosphorodiamidate Piperidino Oligomers By: Ghosh, Atanu; et al Organic Letters (2023), 25(6), 901-906.
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Scheme 46 (1 Reaction)

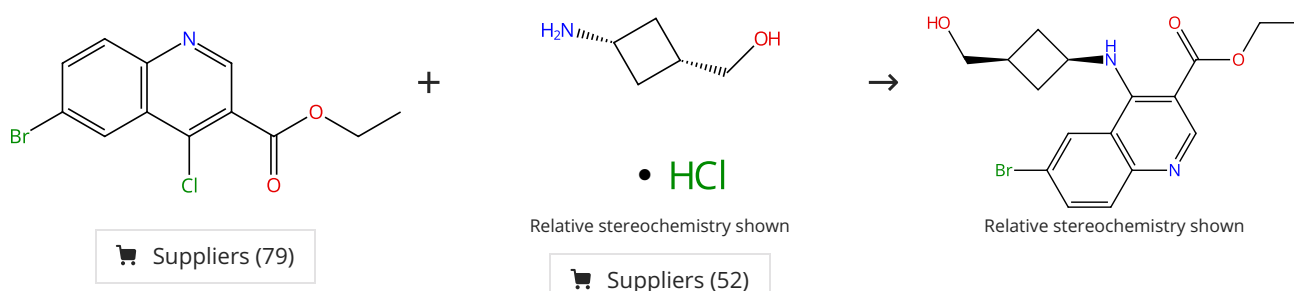
Steps: 1 Yield: 100%



31-614-CAS-41584179 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: Dimethylformamide; rt → 80 °C; overnight, 80 °C Experimental Protocols	Discovery of Novel Nonpeptidic and Noncovalent Small Molecule 3CL^{Pro} Inhibitors as anti-SARS-CoV-2 Drug Candidate By: Jiang, Zhidong; et al Journal of Medicinal Chemistry (2024), 67(15), 12760-12783.
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Scheme 47 (1 Reaction)

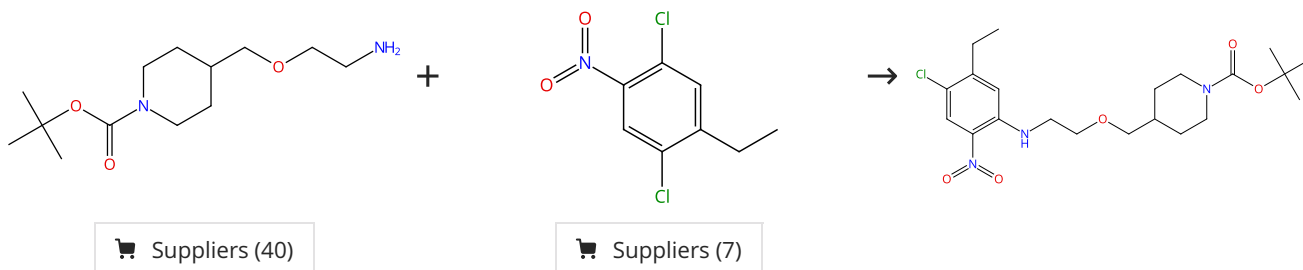
Steps: 1 Yield: 100%



31-614-CAS-39405174	Steps: 1 Yield: 100%	Identification of Novel, Selective Ataxia-Telangiectasia Mutated Kinase Inhibitors with the Ability to Penetrate the Blood-Brain Barrier: The Discovery of AZD1390
1.1 Reagents: Diisopropylethylamine Solvents: Dimethylformamide; 5 - 1080 min, 50 - 100 °C		By: Pike, Kurt G.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2024), 67(4), 3090-3111.

Scheme 48 (1 Reaction)

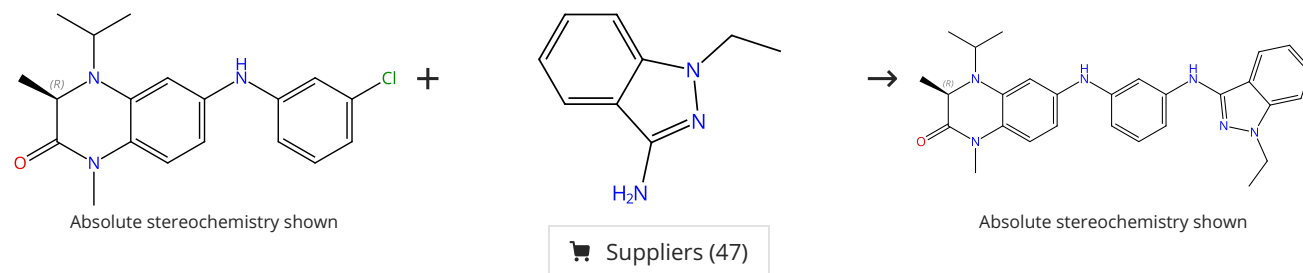
Steps: 1 Yield: 100%



31-614-CAS-36828770	Steps: 1 Yield: 100%	Acetyl-Click Screening Platform Identifies Small-Molecule Inhibitors of Histone Acetyltransferase 1 (HAT1)
1.1 Solvents: Dimethyl sulfoxide; 7 min, 190 °C		By: Gaddameedi, Jitender D.; et al
Experimental Protocols		Journal of Medicinal Chemistry (2023), 66(8), 5774-5801.

Scheme 49 (1 Reaction)

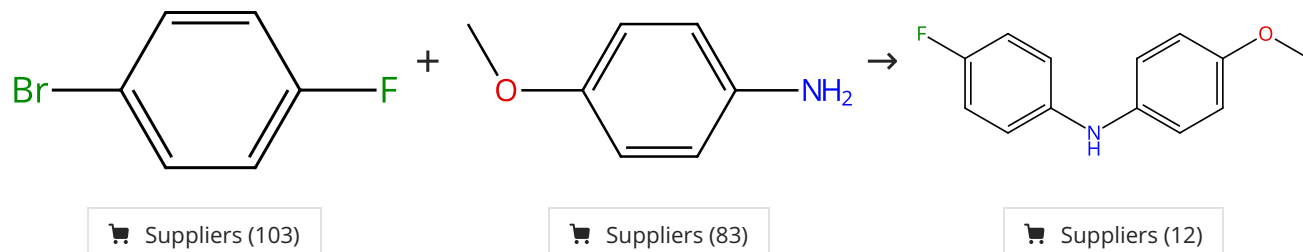
Steps: 1 Yield: 100%



31-614-CAS-44316179	Steps: 1 Yield: 100%	Design of Selective BRD4 Inhibitors for the Treatment of Autosomal Dominant Polycystic Kidney Disease
1.1 Reagents: Sodium <i>tert</i> -butoxide Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos Solvents: Toluene; 8 h, rt → 100 °C		By: Yang, Yueyue; et al
Experimental Protocols		Journal of Medicinal Chemistry (2025), 68(5), 5257-5274.

Scheme 50 (1 Reaction)

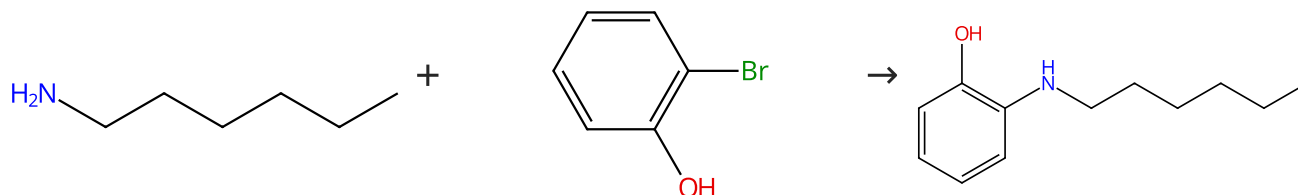
Steps: 1 Yield: 100%



31-614-CAS-46330475	Steps: 1 Yield: 100%	Room-Temperature Copper Cross-Coupling Reactions of Anilines with Aryl Bromides By: Dueker, Jonas; et al Organic Letters (2025), 27(22), 5566-5571.
1.1 Reagents: Potassium carbonate Catalysts: Cuprous iodide, 2-Pyridinecarboxamide, <i>N</i> -(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)-1,6-dihydro-6-oxo- Solvents: Ethanol, Methanol; 21 h, 60 °C		
Experimental Protocols		

Scheme 51 (1 Reaction)

Steps: 1 Yield: 100%



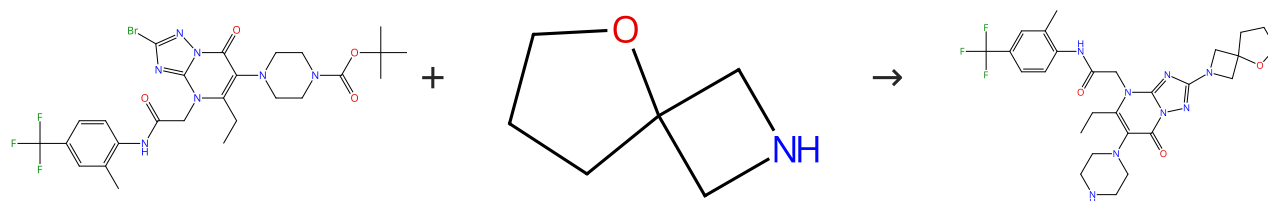
Suppliers (71)

Suppliers (82)

31-614-CAS-46331796	Steps: 1 Yield: 100%	Efficient Aminations of Aryl Halides by a Cu(II) Catalyst By: Pierson, Christina N.; et al Journal of the American Chemical Society (2025), 147(24), 20939-20946.
1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (CuBr ₂), <i>N</i> ¹ , <i>N</i> ² -Bis(2,5-dimethyl-1 <i>H</i> -pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 24 h, 100 °C		
Experimental Protocols		

Scheme 52 (1 Reaction)

Steps: 1 Yield: 100%

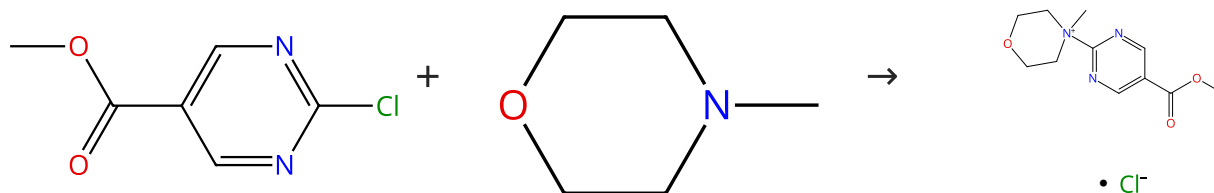


Suppliers (52)

31-614-CAS-42835784	Steps: 1 Yield: 100%	Design, synthesis, and structure-activity relationship studies of triazolo-pyrimidine derivatives as WRN inhibitors for the treatment of MSI tumors By: Sui, Qibang; et al European Journal of Medicinal Chemistry (2025), 282, 117039.
1.1 Reagents: Potassium acetate Solvents: Dimethyl sulfoxide; 12 h, 120 °C		
1.2 Reagents: Trifluoroacetic acid Solvents: Methanol, Dichloromethane; 10 min, rt		
Experimental Protocols		

Scheme 53 (1 Reaction)

Steps: 1 Yield: 100%



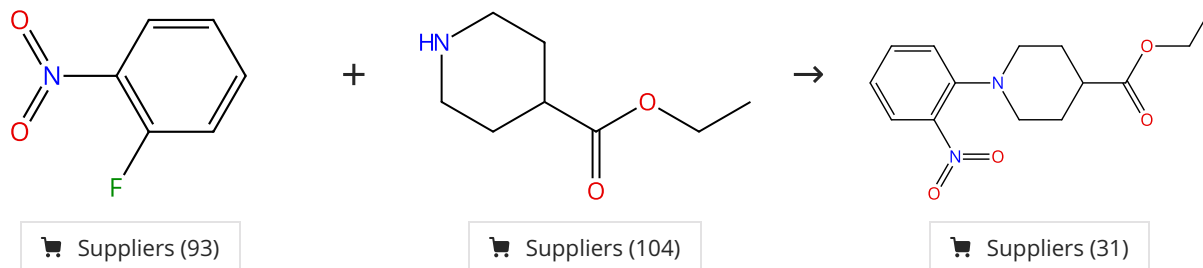
Suppliers (71)

Suppliers (81)

31-614-CAS-40486505 Steps: 1 Yield: 100% 1.1 Solvents: Tetrahydrofuran; 30 min, 25 °C Experimental Protocols	Selective Arylation of RNA 2'-OH Groups via S_NAr Reaction with Trialkylammonium Heterocycles By: Chatterjee, Sayantan; et al Angewandte Chemie, International Edition (2024), 63(25), e202403496.
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Scheme 54 (1 Reaction)

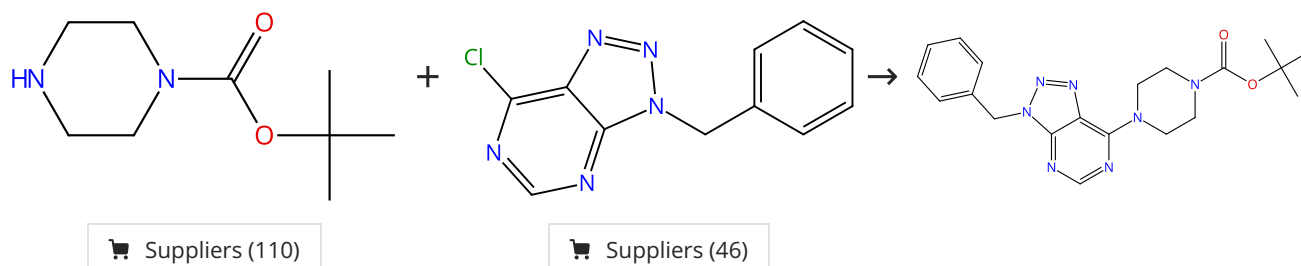
Steps: 1 Yield: 100%



31-614-CAS-36846919 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; 3 h, 90 °C Experimental Protocols	Functionalization of 2,4-Dichloropyrimidines by 2,2,6,6-Tetramethylpiperidyl Zinc Base Enables Modular Synthesis of Antimalarial Diaminopyrimidine P218 and Analogues By: Meirelles, Matheus A.; et al Journal of Organic Chemistry (2023), 88(13), 9475-9487.
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Scheme 55 (1 Reaction)

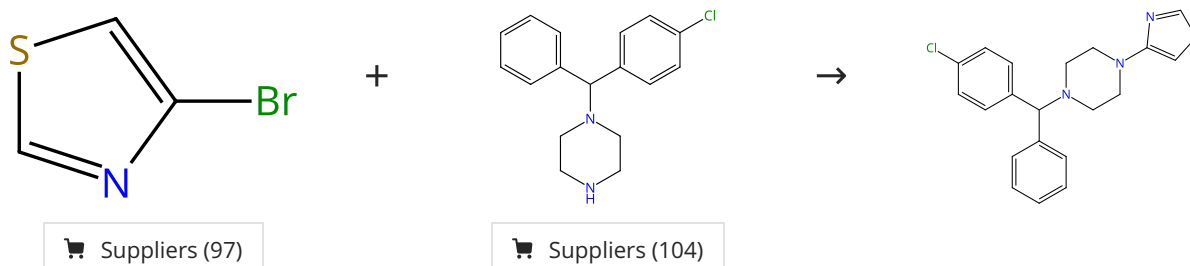
Steps: 1 Yield: 100%



31-614-CAS-44576992 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine Solvents: Ethanol; 1 h, rt 1.2 Reagents: Sodium chloride Solvents: Water	Design of Benzyl-triazolopyrimidine-Based NADPH Oxidase Inhibitors Leads to the Discovery of a Potent Dual Covalent N OX2/MAOB Inhibitor By: Noce, Beatrice; et al Journal of Medicinal Chemistry (2025), 68(6), 6292-6311.
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Scheme 56 (1 Reaction)

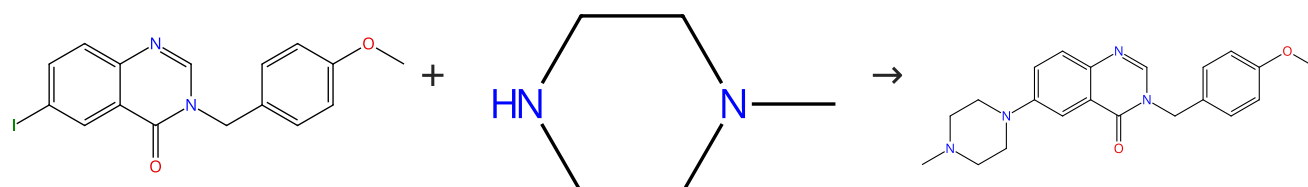
Steps: 1 Yield: 100%



31-614-CAS-35422554 Steps: 1 Yield: 100% 1.1 Reagents: Sodium trimethylsilanolate Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC¹]]phosphine-κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (<i>SP</i>-4-2)- Solvents: Tetrahydrofuran; 3 h, 50 °C Experimental Protocols	Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines By: Reichert, Elaine C.; et al Journal of the American Chemical Society (2023), 145(6), 3323-3329.
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Scheme 57 (1 Reaction)

Steps: 1 Yield: 100%



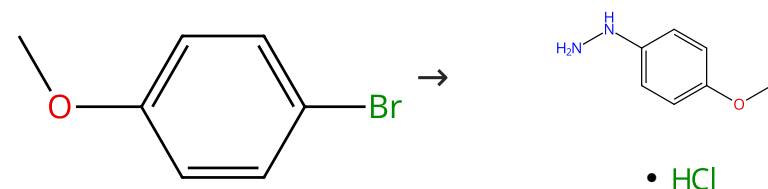
Supplier (1)

Suppliers (106)

31-614-CAS-38625320 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1'-diphenylphosphine] Solvents: 1,4-Dioxane; 24 h, rt → 90 °C Experimental Protocols	Fragment-Based Screening Identifies New Quinazolinone-Based Inositol Hexakisphosphate Kinase (IP6K) Inhibitors By: Heitmann, Tyler; et al ACS Medicinal Chemistry Letters (2023), 14(12), 1760-1766.
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Scheme 58 (1 Reaction)

Steps: 1 Yield: 100%



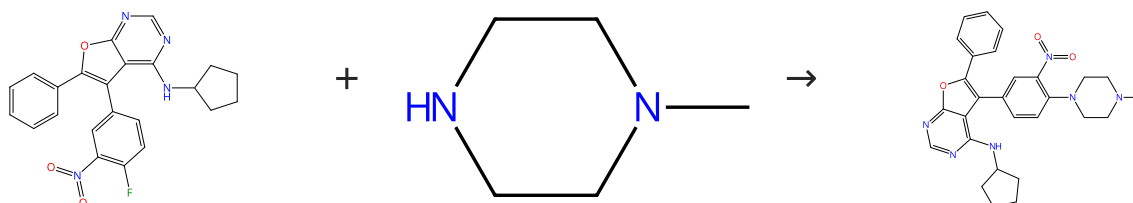
Suppliers (71)

Suppliers (93)

31-614-CAS-46331834 Steps: 1 Yield: 100% 1.1 Reagents: Hydrazine, Tripotassium phosphate Catalysts: Copper bromide (Cu Br₂), <i>N</i>¹,<i>N</i>²-Bis(2,5-dimethyl-1<i>H</i>-pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 24 h, 65 °C 1.2 Reagents: Hydrochloric acid Solvents: Water Experimental Protocols	Efficient Aminations of Aryl Halides by a Cu(II) Catalyst By: Pierson, Christina N.; et al Journal of the American Chemical Society (2025), 147(24), 20939-20946.
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Scheme 59 (1 Reaction)

Steps: 1 Yield: 100%

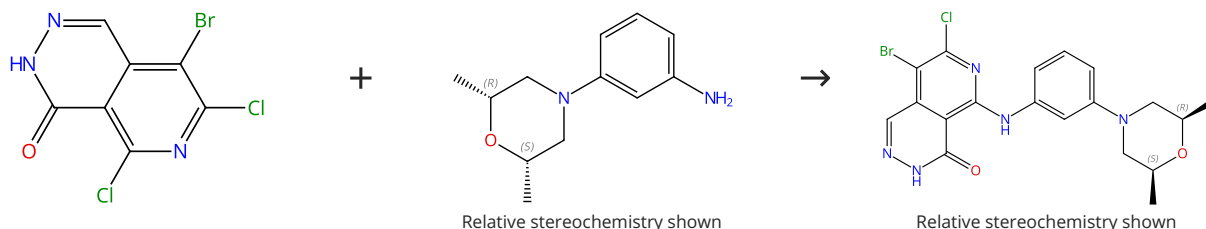


Suppliers (106)

31-614-CAS-35651046	Steps: 1 Yield: 100%	Development of Furanopyrimidine-Based Orally Active Third-Generation EGFR Inhibitors for the Treatment of Non-Small Cell Lung Cancer
1.1 Reagents: Triethylamine Solvents: 1,4-Dioxane; 1 h, reflux		By: Li, Mu-Chun; et al
Experimental Protocols		Journal of Medicinal Chemistry (2023), 66(4), 2566-2588.

Scheme 60 (1 Reaction)

Steps: 1 Yield: 100%

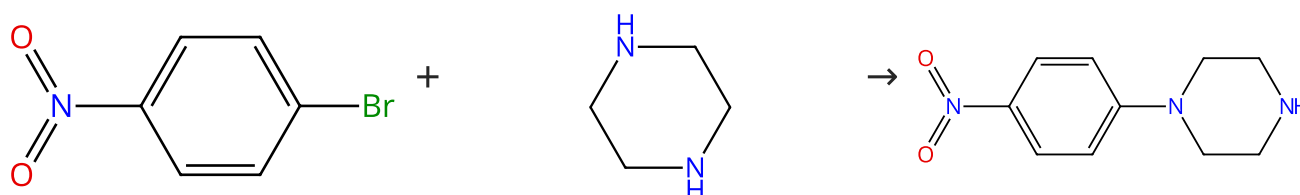


Suppliers (7)

31-614-CAS-41130650	Steps: 1 Yield: 100%	Optimization of Novel Pyrido-pyridazinone Derivatives as FER Tyrosine Kinase Inhibitors, Leading to the Potent DS08701581
1.1 Solvents: <i>N</i> -Methyl-2-pyrrolidone; 1 h, 160 °C		By: Taniguchi, Toru; et al
1.2 Reagents: Sodium chloride Solvents: Water		ACS Medicinal Chemistry Letters (2024), 15(7), 1010-1016.
Experimental Protocols		

Scheme 61 (1 Reaction)

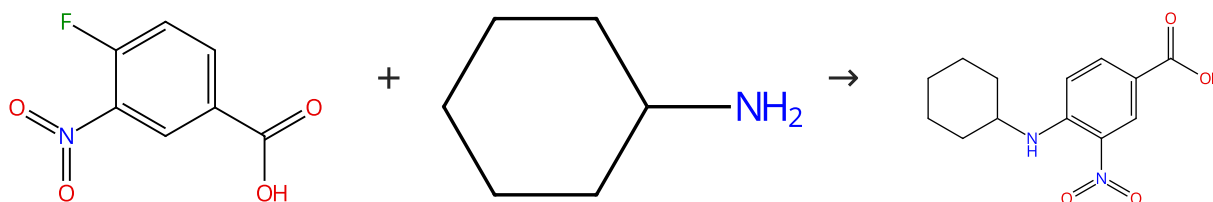
Steps: 1 Yield: 100%



31-614-CAS-37689975	Steps: 1 Yield: 100%	Design, synthesis, and bioactivity evaluation of novel amide/sulfonamide derivatives as potential anti-inflammatory agents against acute lung injury and ulcerative colitis
1.1 Solvents: Dimethyl sulfoxide; 4 h, 80 °C		By: Chen, Pan; et al
Experimental Protocols		European Journal of Medicinal Chemistry (2023), 259, 115706.

Scheme 62 (1 Reaction)

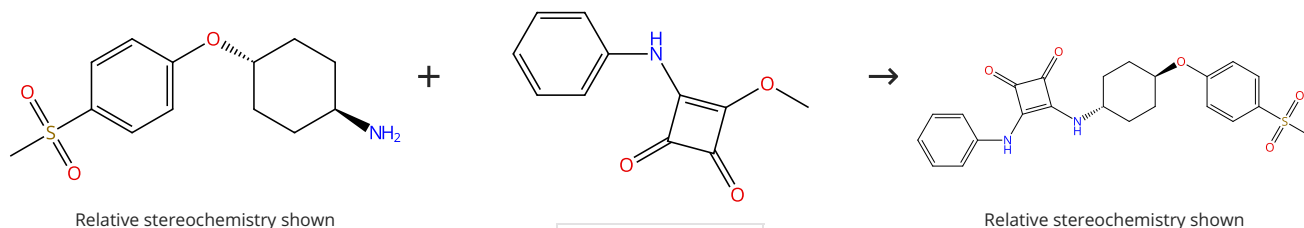
Steps: 1 Yield: 100%



31-614-CAS-44269144	Steps: 1 Yield: 100%	Oxazole-Based Ferroptosis Inhibitors with Promising Properties to Treat Central Nervous System Diseases
1.1 Reagents: Diisopropylethylamine Solvents: Acetonitrile; 72 - 96 h, rt		By: Scarpellini, Camilla; et al
Experimental Protocols		Journal of Medicinal Chemistry (2025), 68(4), 4908-4928.

Scheme 63 (1 Reaction)

Steps: 1 Yield: 100%



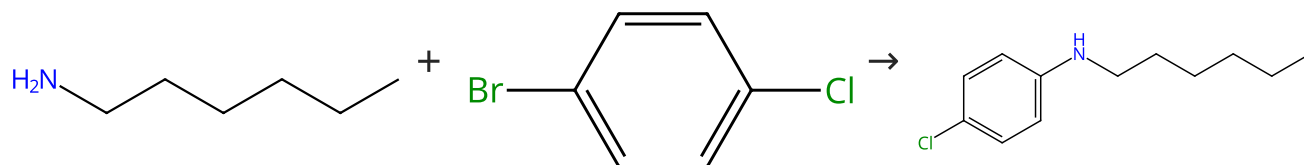
Supplier (1)

Suppliers (3)

31-614-CAS-41838115	Steps: 1 Yield: 100%	Discovery of N-Aryl-N'-[4-(aryloxy)cyclohexyl]squaramide-Based Inhibitors of LXR/SREBP-1c Signaling Pathway Ameliorating Steatotic Liver Disease: Navigating the Role of SIRT6 Activation
1.1 Solvents: Ethanol; 17 h, rt		By: Nguyen, Long Huu; et al
Experimental Protocols		Journal of Medicinal Chemistry (2024), 67(19), 17608-17628.

Scheme 64 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (71)

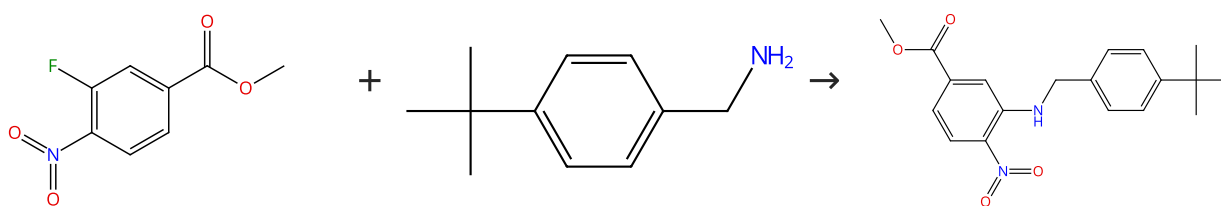
Suppliers (83)

Suppliers (11)

31-614-CAS-46331785	Steps: 1 Yield: 100%	Efficient Aminations of Aryl Halides by a Cu(II) Catalyst
1.1 Reagents: Tripotassium phosphate Catalysts: Copper bromide (CuBr ₂), N ¹ ,N ² -Bis(2,5-dimethyl-1H-pyrrol-1-yl)ethanediamide Solvents: Dimethyl sulfoxide; 24 h, 100 °C		By: Pierson, Christina N.; et al
Experimental Protocols		Journal of the American Chemical Society (2025), 147(24), 20939-20946.

Scheme 65 (1 Reaction)

Steps: 1 Yield: 100%



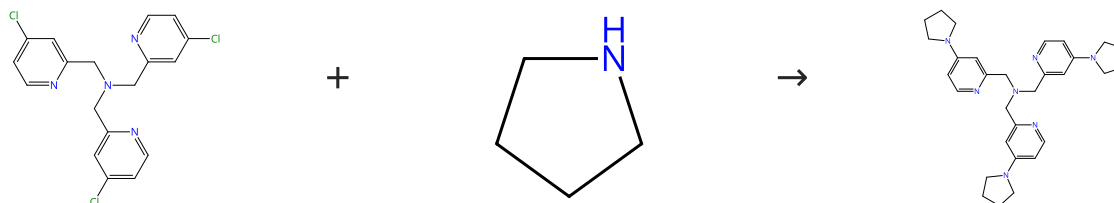
Suppliers (80)

Suppliers (79)

31-614-CAS-36515417	Steps: 1 Yield: 100%	Discovery of FXR/PPARγ dual partial agonist
1.1 Reagents: Triethylamine Solvents: Dimethylformamide; rt; 2 h, 80 °C		By: Yamashita, Yukiko; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Bioorganic & Medicinal Chemistry (2023), 85, 117238.
Experimental Protocols		

Scheme 66 (1 Reaction)

Steps: 1 Yield: 100%

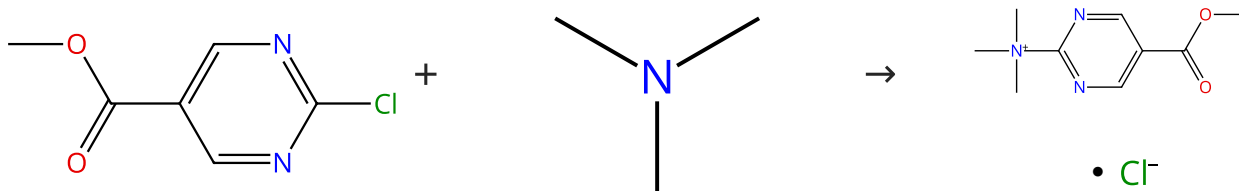


Suppliers (75)

31-614-CAS-41445773	Steps: 1 Yield: 100%	Electrochemically Mediated Atom Transfer Radical Polymerization Driven by Alternating Current
1.1 Reagents: Sodium hydroxide Solvents: Water; 5 min, rt; 18 h, 120 °C		By: De Bon, Francesco; et al
Experimental Protocols		Angewandte Chemie, International Edition (2024), 63(29), e202406484.

Scheme 67 (1 Reaction)

Steps: 1 Yield: 100%



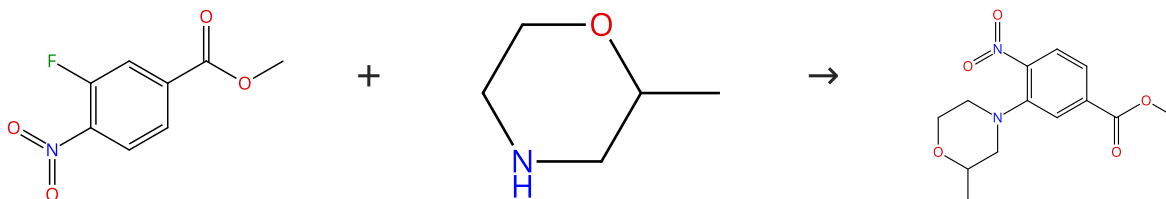
Suppliers (71)

Suppliers (135)

31-614-CAS-40486508	Steps: 1 Yield: 100%	Selective Arylation of RNA 2'-OH Groups via S_NAr Reaction with Trialkylammonium Heterocycles
1.1 Solvents: Tetrahydrofuran; 30 min, 25 °C		By: Chatterjee, Sayantan; et al
Experimental Protocols		Angewandte Chemie, International Edition (2024), 63(25), e202403496.

Scheme 68 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (80)

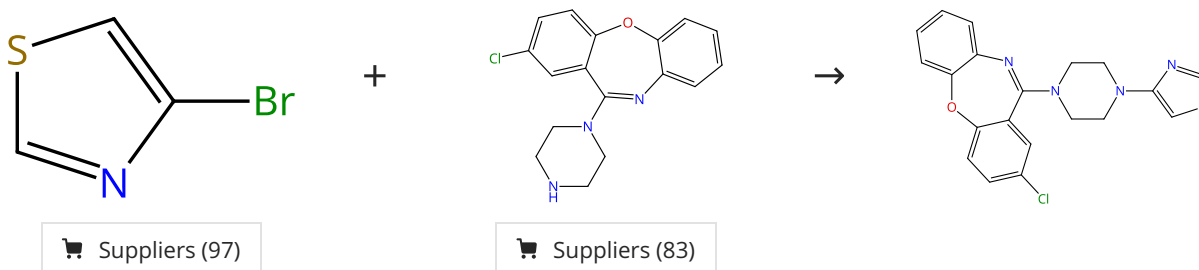
Suppliers (80)

Suppliers (2)

31-614-CAS-36512927 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; 15 h, 60 °C Experimental Protocols	New Dual Inhibitors of Bacterial Topoisomerases with Broad-Spectrum Antibacterial Activity and In Vivo Efficacy against Vancomycin-Intermediate Staphylococcus aureus By: Durcik, Martina; et al Journal of Medicinal Chemistry (2023), 66(6), 3968-3994.
31-614-CAS-40997757 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Acetonitrile; overnight, 60 °C Experimental Protocols	ATP-competitive inhibitors of human DNA topoisomerase IIα with improved antiproliferative activity based on N-phenylpyrrolamide scaffold By: Skok, Ziga; et al European Journal of Medicinal Chemistry (2023), 249, 115116.

Scheme 69 (1 Reaction)

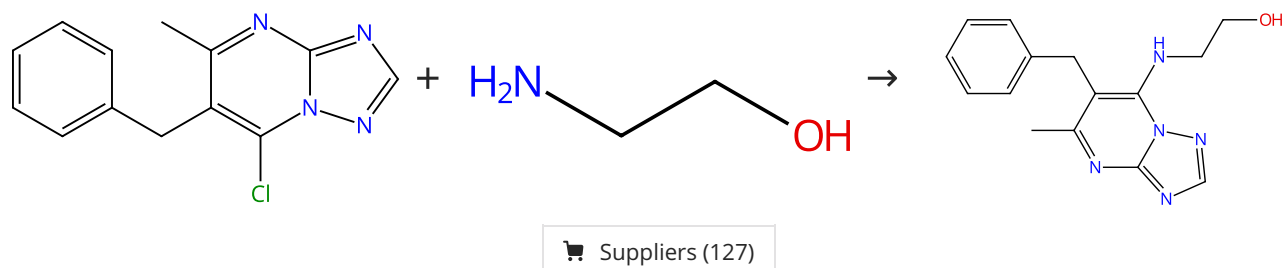
Steps: 1 Yield: 100%



31-614-CAS-35422543 Steps: 1 Yield: 100% 1.1 Reagents: Sodium trimethylsilanolate Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC¹]]phosphine-κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (S^R-4-2)- Solvents: Tetrahydrofuran; 3 h, 50 °C Experimental Protocols	Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines By: Reichert, Elaine C.; et al Journal of the American Chemical Society (2023), 145(6), 3323-3329.
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Scheme 70 (1 Reaction)

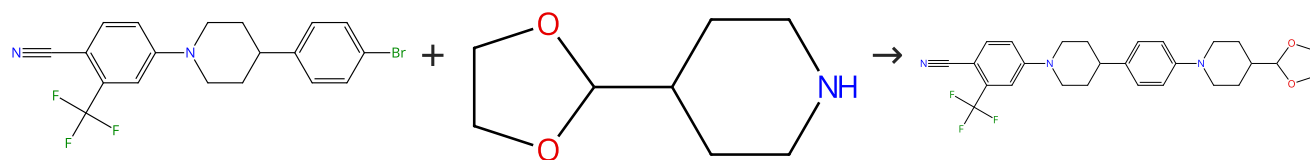
Steps: 1 Yield: 100%



31-614-CAS-41352674 Steps: 1 Yield: 100% 1.1 Solvents: Ethanol; 6 h, rt Experimental Protocols	[1,2,4]Triazolo[1,5-a]pyrimidine derivatives: Structure-activity relationship study leading to highly selective ENPP1 inhibitors By: Kawaguchi, Mitsuyasu; et al Bioorganic & Medicinal Chemistry Letters (2024), 110, 129820.
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Scheme 71 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (52)

31-614-CAS-41378152

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl] dicyclohexylphosphine, [2'-(Amino- κM)[1,1'-biphenyl]-2-yl- κC] [[2',6'-bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine- κP](methanesulfonato- κO)palladium
Solvents: 1,4-Dioxane; 2 h, 100 °C

Experimental Protocols

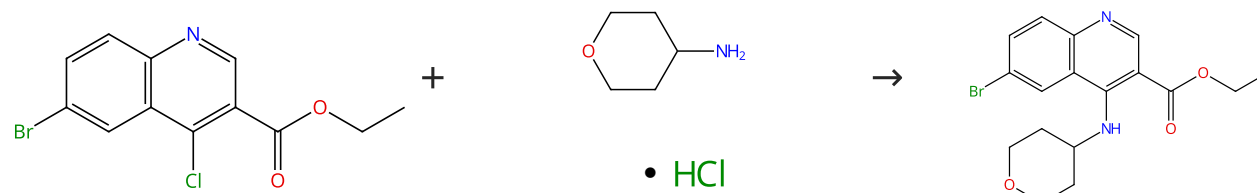
Discovery of a Series of Orally Bioavailable Androgen Receptor Degraders for the Treatment of Prostate Cancer

By: Bagal, Sharan K.; et al

Journal of Medicinal Chemistry (2024), 67(14), 11732-11750.

Scheme 72 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (79)

Suppliers (78)

Suppliers (9)

31-614-CAS-39405406

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Diisopropylethylamine
Solvents: Dimethylformamide; 5 - 1080 min, 50 - 100 °C

Experimental Protocols

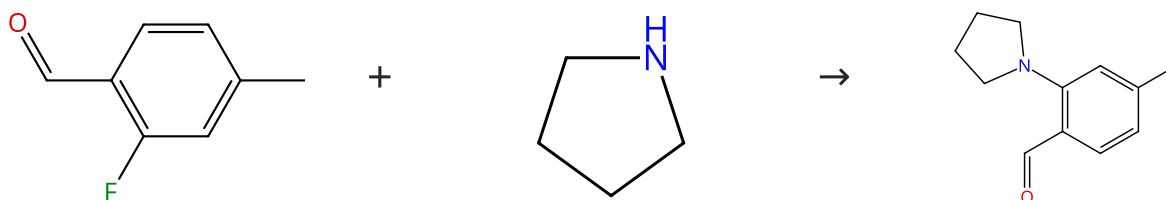
Identification of Novel, Selective Ataxia-Telangiectasia Mutated Kinase Inhibitors with the Ability to Penetrate the Blood-Brain Barrier: The Discovery of AZD1390

By: Pike, Kurt G.; et al

Journal of Medicinal Chemistry (2024), 67(4), 3090-3111.

Scheme 73 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (68)

Suppliers (75)

Suppliers (4)

31-614-CAS-45702640

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 - 4 h, rt → reflux; reflux → rt
 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

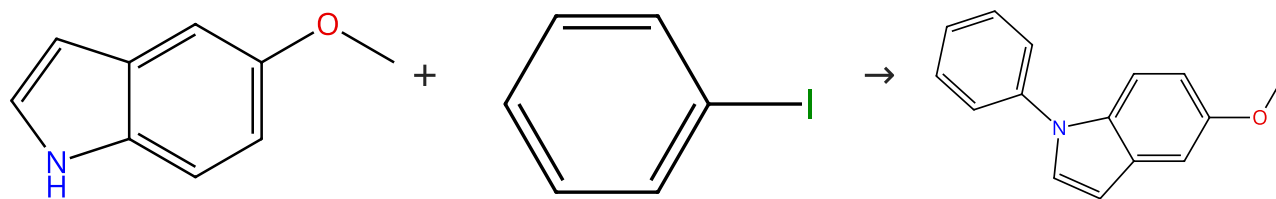
Construction of spirotetrahydroquinolines fused with medium-sized rings via TBHP/base-promoted spirocyclization reactions

By: He, Peng; et al

Organic Chemistry Frontiers (2025), 12(19), 5346-5350.

Scheme 74 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (105)

Suppliers (96)

Suppliers (7)

31-614-CAS-39303569

Steps: 1 Yield: 100%

Tosylazide as N1-Synthon: Iron-Catalyzed Nitrogenative Dimerization of Indoles to p-Bisindolopyrazine Derivatives

By: Li, Jianan; et al

Organic Letters (2024), 26(5), 1046-1050.

1.1 Reagents: Cesium carbonate

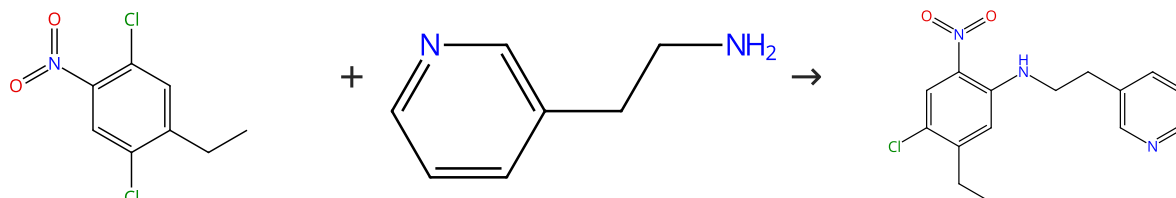
Catalysts: Cuprous iodide

Solvents: Dimethylformamide; overnight, 120 °C

Experimental Protocols

Scheme 75 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (7)

Suppliers (74)

31-614-CAS-36828639

Steps: 1 Yield: 100%

Acetyl-Click Screening Platform Identifies Small-Molecule Inhibitors of Histone Acetyltransferase 1 (HAT1)

By: Gaddameedi, Jitender D.; et al

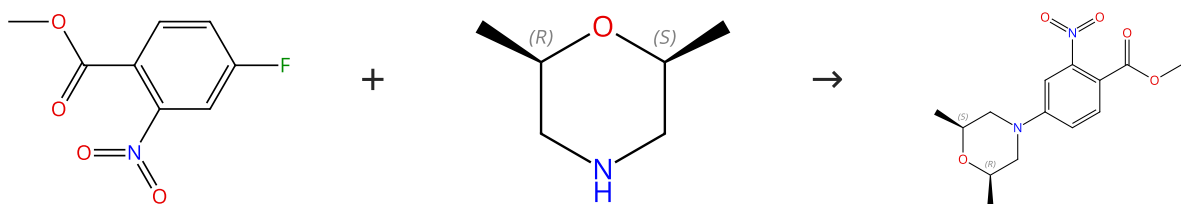
Journal of Medicinal Chemistry (2023), 66(8), 5774-5801.

1.1 Solvents: Dimethyl sulfoxide; 15 min, 190 °C

Experimental Protocols

Scheme 76 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (57)

Relative stereochemistry shown

Suppliers (97)

Relative stereochemistry shown

31-614-CAS-35431833

Steps: 1 Yield: 100%

Potent Long-Acting Inhibitors Targeting the HIV-1 Capsid Based on a Versatile Quinazolin-4-one Scaffold

By: Gillis, Eric P.; et al

Journal of Medicinal Chemistry (2023), 66(3), 1941-1954.

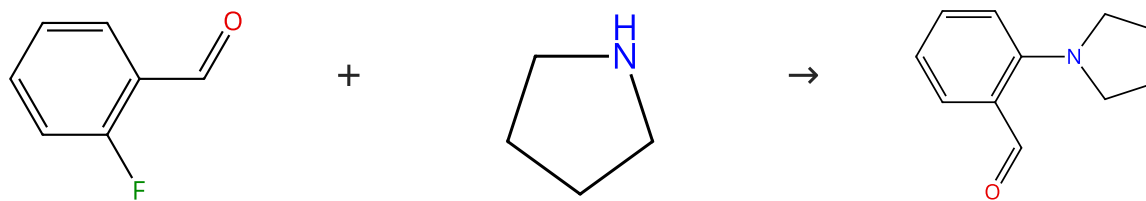
1.1 Reagents: Potassium carbonate

Solvents: Dimethylformamide; 108 h, rt

Experimental Protocols

Scheme 77 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (75)

Suppliers (58)

31-614-CAS-45702636

Steps: 1 Yield: 100%

Construction of spirotetrahydroquinolines fused with medium-sized rings via TBHP/base-promoted spirocyclization reactions

By: He, Peng; et al

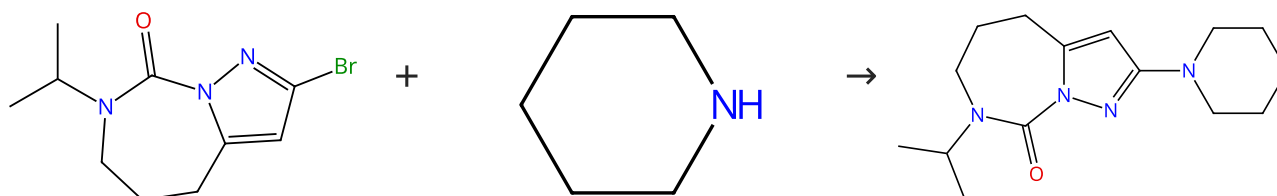
Organic Chemistry Frontiers (2025), 12(19), 5346-5350.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 - 4 h, rt → reflux; reflux → rt
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 78 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (54)

31-614-CAS-35422566

Steps: 1 Yield: 100%

Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines

By: Reichert, Elaine C.; et al

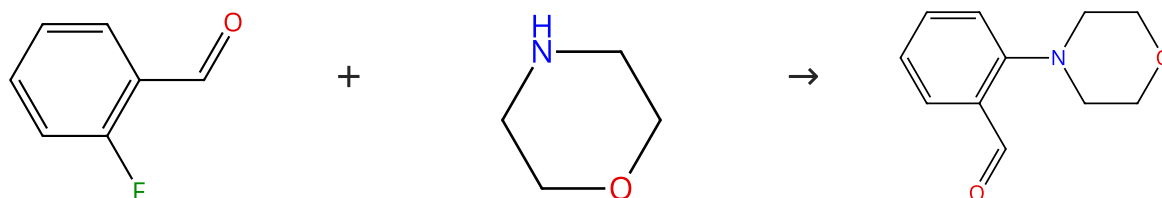
Journal of the American Chemical Society (2023), 145(6), 3323-3329.

- 1.1 **Reagents:** Sodium trimethylsilanolate
Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC¹]]phosphine-κP[4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (*SP*-4-2)-
Solvents: Tetrahydrofuran; rt → 90 °C; 3 h, 90 °C

Experimental Protocols

Scheme 79 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (88)

Suppliers (71)

31-614-CAS-45702639

Steps: 1 Yield: 100%

Construction of spirotetrahydroquinolines fused with medium-sized rings via TBHP/base-promoted spirocyclization reactions

By: He, Peng; et al

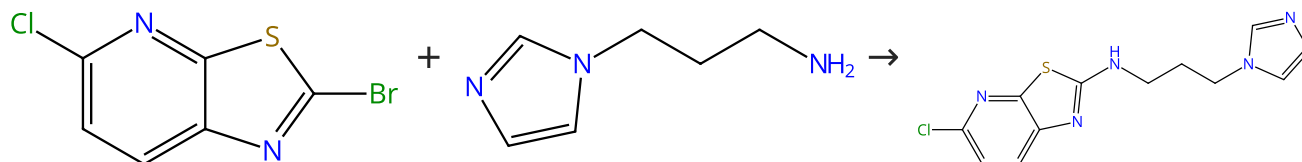
Organic Chemistry Frontiers (2025), 12(19), 5346-5350.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 - 4 h, rt → reflux; reflux → rt
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 80 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (46)

Suppliers (86)

31-614-CAS-37043346

Steps: 1 Yield: 100%

Identification and structure-activity relationships for a series of N, N-disubstituted 2-aminobenzothiazoles as potent inhibitors of *S. aureus*

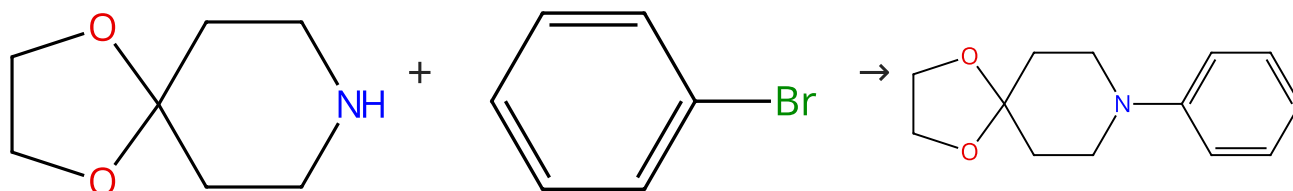
By: Cao, Feng; et al

Bioorganic & Medicinal Chemistry Letters (2023), 89, 129301.

1.1 Reagents: Potassium carbonate
Solvents: 1,4-Dioxane; overnight, 100 °C

Scheme 81 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (57)

Suppliers (80)

Suppliers (16)

31-614-CAS-41624313

Steps: 1 Yield: 100%

Discovery of DS-1093a: An oral hypoxia-inducible factor prolyl hydroxylase inhibitor for the treatment of renal anemia

By: Tanaka, Naoki; et al

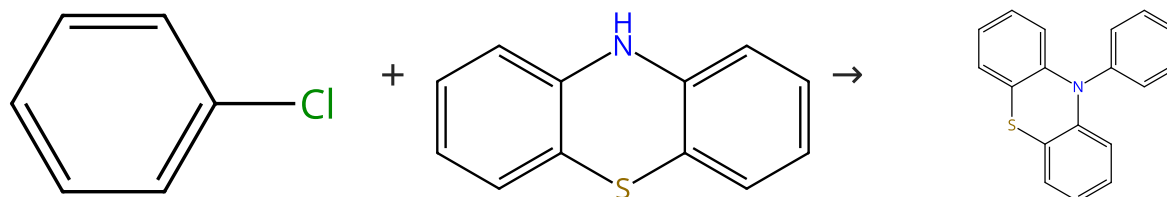
Bioorganic & Medicinal Chemistry Letters (2024), 111, 129891.

1.1 Reagents: Sodium *tert*-butoxide
Catalysts: Palladium, tris[μ-[(1,2-η:4,5-η)-(1*E*,4*E*)-1,5-diphenyl-1,4-pentadien-3-one]]di-, compd. with trichloromethane (1:1), 2-Dicyclohexylphosphino-2',6'-dimethoxybiphenyl
Solvents: Toluene; 3 h, rt → reflux

Experimental Protocols

Scheme 82 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (127)

Suppliers (105)

Suppliers (60)

31-614-CAS-44934216

Steps: 1 Yield: 100%

Phenothiazine Sulfoxides as Active Photocatalysts for the Synthesis of γ-Lactones

By: Hoelter, Niklas; et al

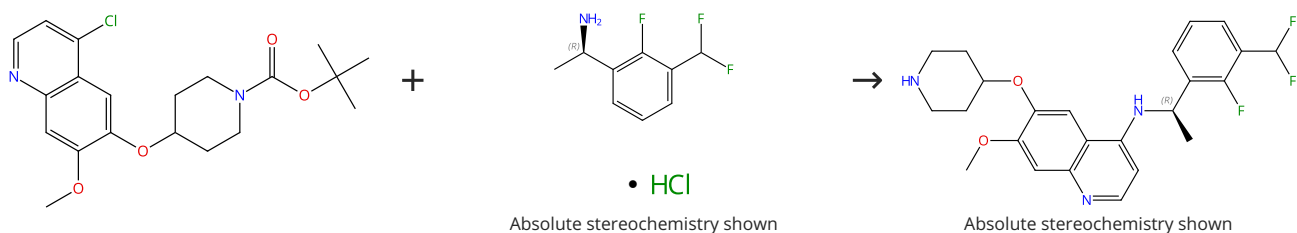
Journal of the American Chemical Society (2025), 147(15), 12908-12916.

1.1 Reagents: Sodium *tert*-butoxide
Catalysts: [2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine, (*SP*-4-3)-[[2',6'-Bis(1-methylethoxy)[1,1'-biphenyl]-2-yl]dicyclohexylphosphine-κ*P*](methanesulfonato-κ*O*)[2'-(methylamino-κ*M*)[1,1'-biphenyl]-2-yl-κ*C*]palladium
Solvents: 1,4-Dioxane; 15 h, 110 °C

Experimental Protocols

Scheme 83 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (46)

31-614-CAS-40791270

Steps: 1 Yield: 100%

Lead Identification of Novel Naphthyridine Derivatives as Potent SOS1 Inhibitors

By: Li, Dongsheng; et al

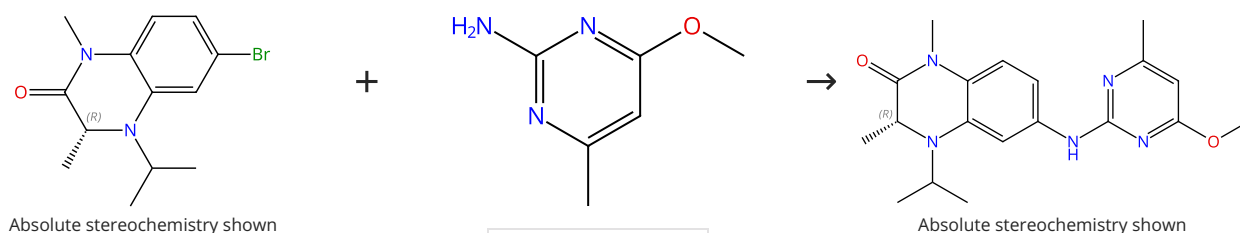
ACS Medicinal Chemistry Letters (2024), 15(6), 958-964.

- 1.1 Reagents: Cesium carbonate
Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos
Solvents: Toluene; overnight, 110 °C
- 1.2 Reagents: Trifluoroacetic acid
Solvents: Dichloromethane; 2 h, rt
- 1.3 Reagents: Ammonium hydroxide; pH 9, 0 °C

Experimental Protocols

Scheme 84 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

Suppliers (68)

31-614-CAS-44316146

Steps: 1 Yield: 100%

Design of Selective BRD4 Inhibitors for the Treatment of Autosomal Dominant Polycystic Kidney Disease

By: Yang, Yueyue; et al

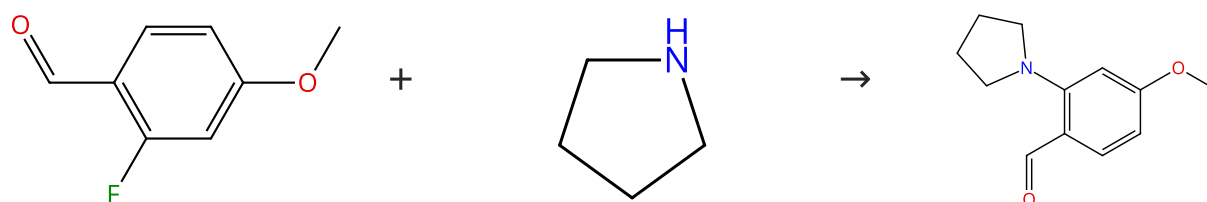
Journal of Medicinal Chemistry (2025), 68(5), 5257-5274.

- 1.1 Reagents: Sodium *tert*-butoxide
Catalysts: Tris(dibenzylideneacetone)dipalladium, X-Phos
Solvents: Toluene; 8 h, rt → 100 °C

Experimental Protocols

Scheme 85 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (88)

Suppliers (75)

Supplier (1)

31-614-CAS-45702638

Steps: 1 Yield: 100%

Construction of spirotrihydroquinolines fused with medium-sized rings via TBHP/base-promoted spirocyclization reactions

By: He, Peng; et al

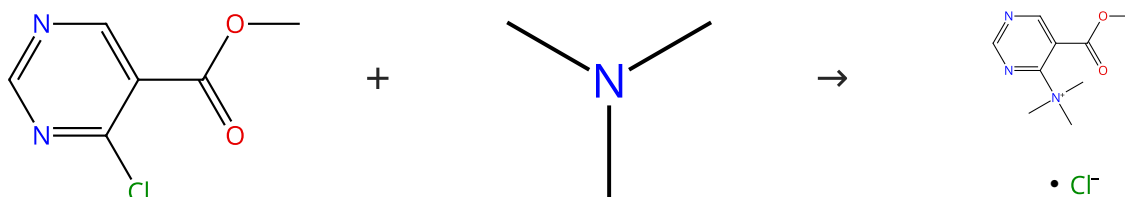
Organic Chemistry Frontiers (2025), 12(19), 5346-5350.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 - 4 h, rt → reflux; reflux → rt
- 1.2 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 86 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (50)

Suppliers (135)

31-614-CAS-40486510

Steps: 1 Yield: 100%

Selective Arylation of RNA 2'-OH Groups via S_NAr Reaction with Trialkylammonium Heterocycles

1.1 Solvents: Tetrahydrofuran; 30 min, 25 °C

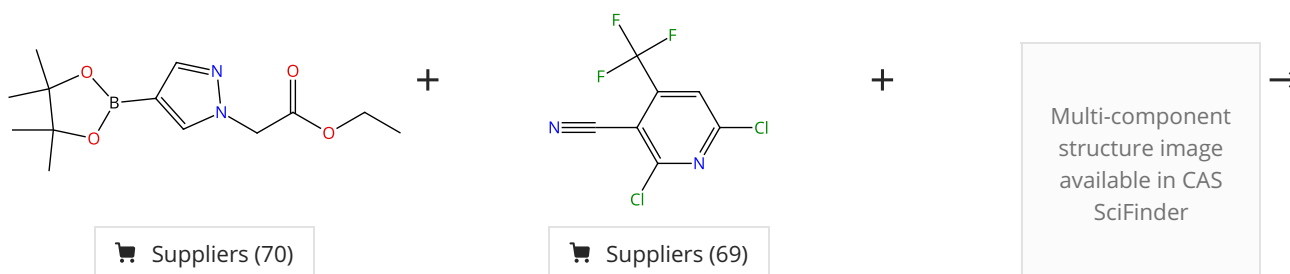
Experimental Protocols

By: Chatterjee, Sayantan; et al

Angewandte Chemie, International Edition (2024), 63(25), e202403496.

Scheme 87 (1 Reaction)

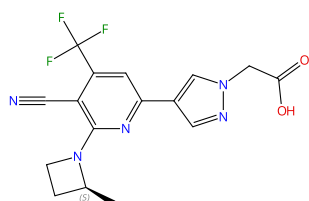
Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (69)

Suppliers (36)



Absolute stereochemistry shown

31-614-CAS-38357811

Steps: 1 Yield: 100%

Identification of LY3522348: A Highly Selective and Orally Efficacious Ketohexokinase (KHK) Inhibitor

1.1 Reagents: Sodium carbonate
Catalysts: Dichlorobis(triphenylphosphine)palladium
Solvents: 1,4-Dioxane, Water; 16.5 h, rt

1.2 rt → 70 °C; 3 h, 70 °C; 70 °C → rt

1.3 Reagents: Sodium hydroxide
Solvents: Water; 15 min, rt

1.4 Reagents: Citric acid
Solvents: 2-Methyltetrahydrofuran, Water; 5 min, rt

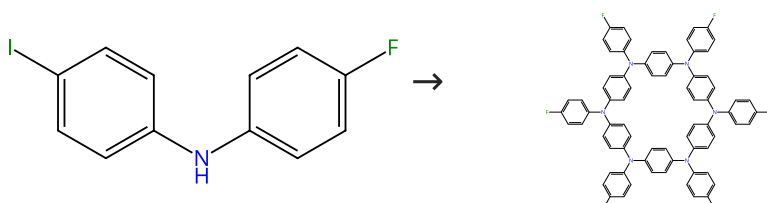
Experimental Protocols

By: Durham, Timothy B.; et al

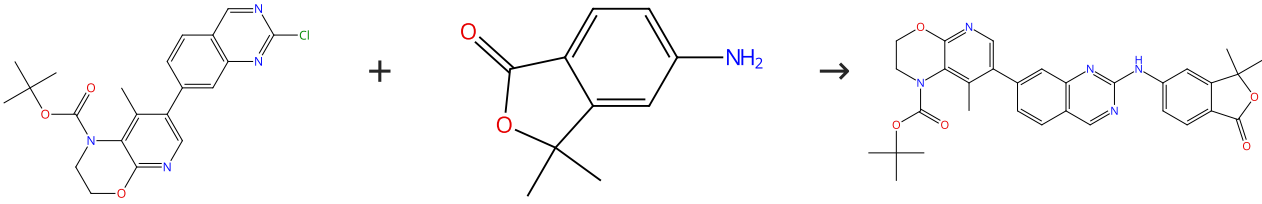
Journal of Medicinal Chemistry (2023), 66(23), 15960-15976.

Scheme 88 (1 Reaction)

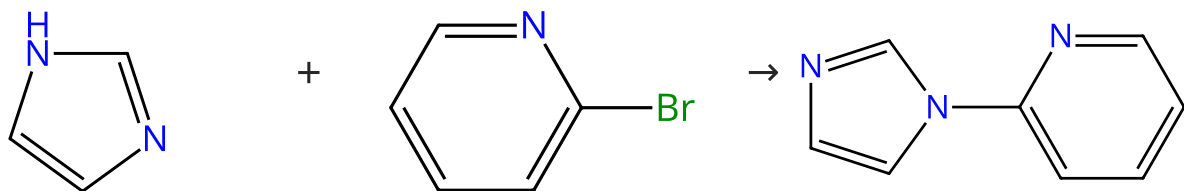
Steps: 1 Yield: 100%



31-614-CAS-41358475 Steps: 1 Yield: 100% 1.1 Reagents: Sodium <i>tert</i>-butoxide Catalysts: X-Phos, (<i>SP</i>-4-3)-[Dicyclohexyl[2',4',6'-tris(1-methyl ethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato-κO) [2'-(methylamino-κN)[1,1'-biphenyl]-2-yl-κC]palladium Solvents: Tetrahydrofuran; 1 h, 40 °C 1.2 2 h, 40 °C 1.3 Reagents: Hydrochloric acid Solvents: Methanol, Water Experimental Protocols	One-Step Catalyst-Transfer Macrocyclization: Expanding the Chemical Space of Azaparacyclophanes By: Ayuso-Carrillo, Josue; et al Journal of the American Chemical Society (2024), 146(24), 16440-16457.
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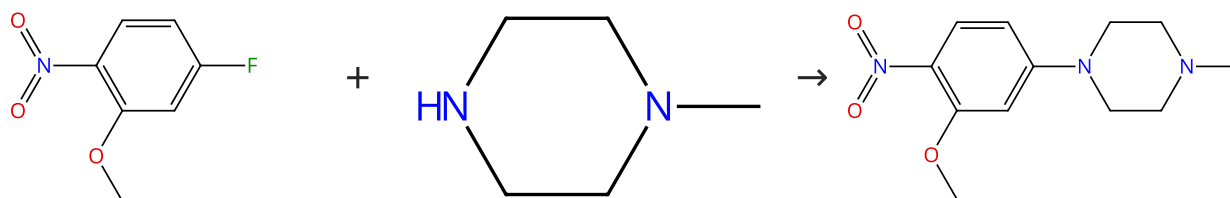
Scheme 89 (1 Reaction) Steps: 1 Yield: 100%  Suppliers (51)
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31-614-CAS-37427271 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: (<i>SP</i>-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-4,5-dichloro-1,3-dihydro-2<i>H</i>-imidazol-2-ylidene]dichloro(2-methylpyridine)palladium Solvents: 1,4-Dioxane; 16 h, 100 °C Experimental Protocols	Discovery of quinazoline HPK1 inhibitors with high cellular potency By: Toure, Momar; et al Bioorganic & Medicinal Chemistry (2023), 92, 117423.
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Scheme 90 (1 Reaction) Steps: 1 Yield: 100%  Suppliers (207) Suppliers (88) Suppliers (45)

31-614-CAS-36663155 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate; 18 h, rt → 190 °C Experimental Protocols	A Hemilabile NHC-Gold Complex and its Application to the Redox Neutral 1,2-Oxyarylation of Feedstock Alkenes By: Scott, Samuel C.; et al Angewandte Chemie, International Edition (2023), 62(23), e202301526.
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Scheme 91 (1 Reaction) Steps: 1 Yield: 100%



Suppliers (91)

Suppliers (106)

Suppliers (54)

31-614-CAS-36680719

Steps: 1 Yield: 100%

Addressing the Osimertinib Resistance Mutation EGFR-L858R/C797S with Reversible Aminopyrimidines

By: Grabe, Tobias; et al

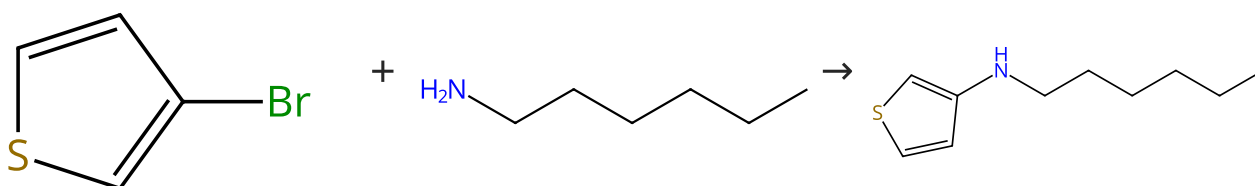
ACS Medicinal Chemistry Letters (2023), 14(5), 591-598.

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 50 °C

Experimental Protocols

Scheme 92 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (82)

Suppliers (71)

Suppliers (6)

31-614-CAS-35422564

Steps: 1 Yield: 100%

Pd-Catalyzed Amination of Base-Sensitive Five-Membered Heteroaryl Halides with Aliphatic Amines

By: Reichert, Elaine C.; et al

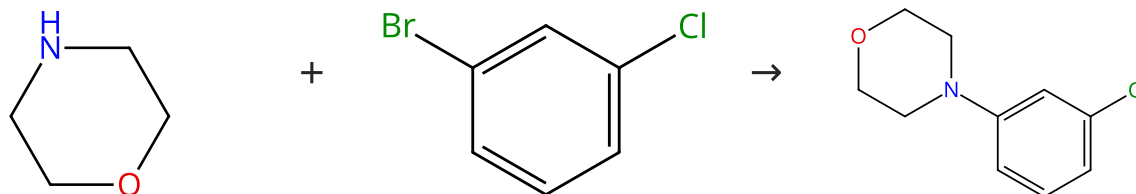
Journal of the American Chemical Society (2023), 145(6), 3323-3329.

1.1 **Reagents:** Sodium trimethylsilanolate
Catalysts: Palladium, bromo[dicyclohexyl[3-(1,1-dimethylethoxy)-6-methoxy-2',6'-bis(1-methylethyl)[1,1'-biphenyl]-2-yl-κC¹]phosphine-κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (*SP*-4-2)-
Solvents: Tetrahydrofuran; 3 h, 70 °C

Experimental Protocols

Scheme 93 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (88)

Suppliers (83)

Suppliers (42)

31-614-CAS-41756977

Steps: 1 Yield: 100%

Copper-Catalyzed Amination of Aryl Chlorides under Mild Reaction Conditions

By: Ai, Han-Jun; et al

Journal of the American Chemical Society (2024), 146(38), 25949-25955.

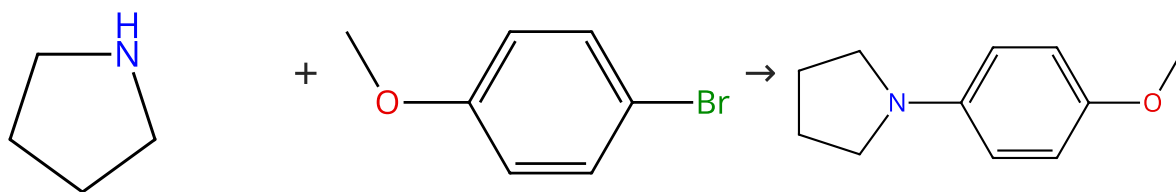
1.1 **Reagents:** Sodium methoxide
Catalysts: Copper bromide (CuBr), 1,2-Benzenediamine, *N*¹-[3',5'-bis(1,1-dimethylethyl)[1,1'-biphenyl]-2-yl]-*N*²-[2-[3,5-bis(1,1-dimethylethyl)phenyl]-1-naphthalenyl]-
Solvents: Dimethyl sulfoxide; 5 min

1.2 2 h, rt

Experimental Protocols

Scheme 94 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (71)

Suppliers (39)

31-614-CAS-46331827

Steps: 1 Yield: 100%

Efficient Aminations of Aryl Halides by a Cu(II) Catalyst

1.1 **Reagents:** Tripotassium phosphate
Catalysts: Copper bromide (CuBr₂), *N*¹,*N*²-Bis(2,5-dimethyl-1*H*-pyrrol-1-yl)ethanediamide
Solvents: Dimethyl sulfoxide; 24 h, 120 °C

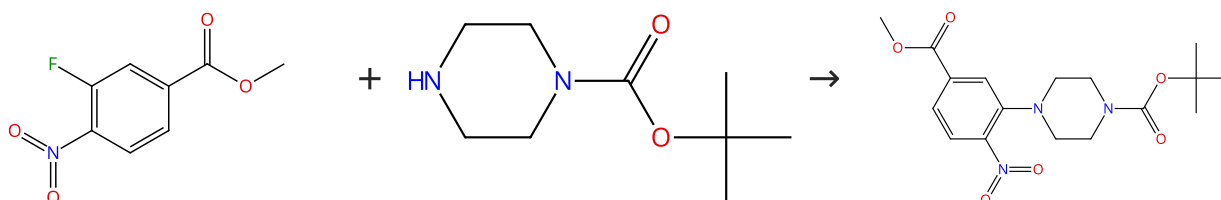
By: Pierson, Christina N.; et al

Journal of the American Chemical Society (2025), 147(24), 20939-20946.

Experimental Protocols

Scheme 95 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

Suppliers (110)

31-614-CAS-36512954

Steps: 1 Yield: 100%

New Dual Inhibitors of Bacterial Topoisomerases with Broad-Spectrum Antibacterial Activity and In Vivo Efficacy against Vancomycin-Intermediate *Staphylococcus aureus*

1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile; 15 h, 60 °C

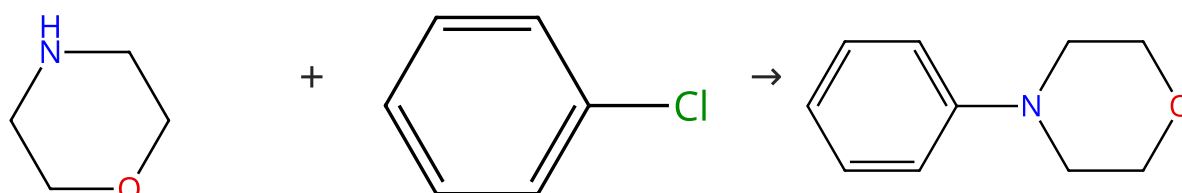
By: Durcik, Martina; et al

Journal of Medicinal Chemistry (2023), 66(6), 3968-3994.

Experimental Protocols

Scheme 96 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (88)

Suppliers (127)

Suppliers (76)

31-614-CAS-42011789

Steps: 1 Yield: 100%

Development of a Deactivation-Resistant Dialkylbiarylphosphine Ligand for Pd-Catalyzed Arylation of Secondary Amines

1.1 **Reagents:** Sodium *tert*-butoxide
Catalysts: 3059853-81-2
Solvents: 1,4-Dioxane; 1 h, 60 °C

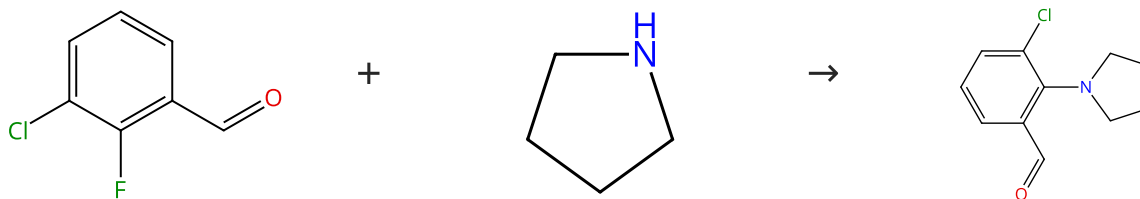
By: Feng, Kaibo; et al

Journal of the American Chemical Society (2024), 146(39), 26609-26615.

Experimental Protocols

Scheme 97 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (87)

Suppliers (75)

Suppliers (6)

31-614-CAS-45702635

Steps: 1 Yield: 100%

Construction of spirotetrahydroquinolines fused with medium-sized rings via TBHP/base-promoted spirocyclization reactions

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 - 4 h, rt → reflux; reflux → rt
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

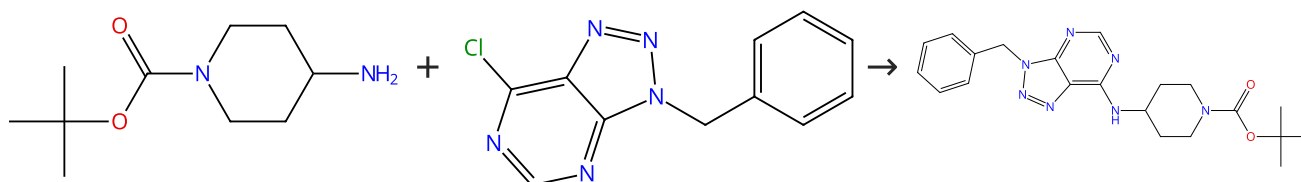
By: He, Peng; et al

Organic Chemistry Frontiers (2025), 12(19), 5346-5350.

Experimental Protocols

Scheme 98 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (101)

Suppliers (46)

31-614-CAS-44576995

Steps: 1 Yield: 100%

Design of Benzyl-triazolopyrimidine-Based NADPH Oxidase Inhibitors Leads to the Discovery of a Potent Dual Covalent N OX2/MAOB Inhibitor

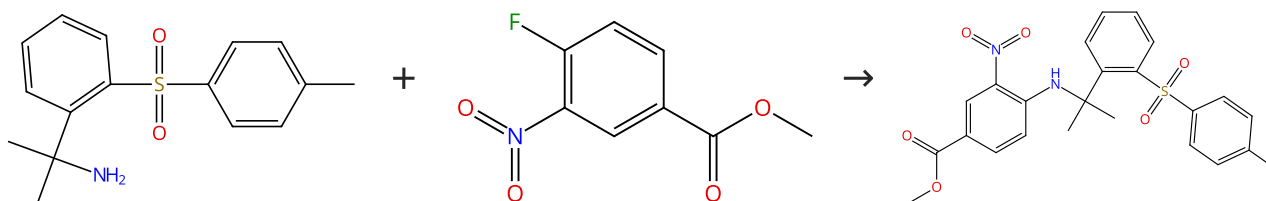
- 1.1 **Reagents:** Triethylamine
Solvents: Ethanol; 1 h, rt
- 1.2 **Reagents:** Sodium chloride

By: Noce, Beatrice; et al

Journal of Medicinal Chemistry (2025), 68(6), 6292-6311.

Scheme 99 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

31-614-CAS-37155500

Steps: 1 Yield: 100%

Dual Copper- and Aldehyde-Catalyzed Transient C-H Sulfonylation of Benzylamines

- 1.1 **Solvents:** Isopropanol; 4 h, 100 °C

By: Higham, Joe I.; et al

Organic Letters (2023), 25(28), 5285-5290.

Experimental Protocols